

Modern Geometry I course notes (Instructor Version)

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“It is not knowledge, but the act of learning, not possession but the act of getting there, which grants the greatest enjoyment.”

—Karl Friedrich Gauss, 1808

Acknowledgements

The materials used in these course notes have been drawn from a variety of sources. The NG worksheets in Section 5 were written by Charles Coppin for a course at the University of Dallas; they are reprinted here with his kind permission. Several of the assignments refer to questions from the book *Experiencing Geometry In Euclidean, Spherical, and Hyperbolic Spaces* by David W. Henderson and Daina Tamina [Henderson & Taimina, 2005]; the statements of these questions are included, with the authors' permission, in Appendix D. Several assignments refer to Euclid's *Elements*, excerpts of which appear in Appendix C in Sir Thomas Heath's English translation [Euclid, 1956]. This translation is in the public domain. Section 11.1 reproduces the 2001 NCTM Standards for School Mathematics [NCTM, 2000], developed by the National Council of Teachers of Mathematics. An excerpt from *Flatland* by Edwin Abbott [Abbott, 1885] appears in Appendix B. *Flatland* is in the public domain.

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Preface

“The royal road to knowledge,
it is easy to express:
to err, and err, and err again,
but less, and less, and less.”

This course is about geometry, but it is also a course about learning to develop and express your own mathematical ideas. It will emphasize ideas and imagination in addition to techniques and calculations. We will be investigating not only the planar Euclidean geometry that you probably studied in high school, but also geometry on spheres, cylinders, cones, pool tables, and other surfaces. We will try to imagine what geometry would be like for someone living on each of these surfaces. We will be studying these geometries not only because they are useful and can tell us surprising things about the world in which we live, but also because they are beautiful and fascinating subjects.

This course may be very different from most other math courses that you have taken. High school and introductory college math courses usually focus on teaching methods of doing computations: mathematics as finding the correct answer. In this course, we are going to focus on another kind of mathematics: mathematics as a way of thinking about and trying to understand the world. We’ll try to understand how people decide what is true, and how they reason about mathematics and geometry in particular. We are going to focus on the process as much as the results of mathematical thought.

Writing is an important part of this process. Contrary to what some other math courses may have led you to believe, it is virtually impossible to do mathematics without writing about it. Writing is a tool for communicating ideas to other people, but it can also be used as a tool for clarifying one’s own ideas. It can be very hard to spot a flaw in a line of reasoning if you haven’t written it down; conversely, writing down a line of reasoning is often the best way to expose any problems that it might have. For these reasons, practicing mathematicians invariably use writing as a vital part of their work, and writing will be an integral part of this course.

This course will require a willingness to invest significant amounts of effort grappling with developing your own ideas. In this course, as in the real world, you will be the ultimate arbiter of what you believe to be true. Deciding for yourself what is true is not easy, but it can be very rewarding.

Introduction for the Instructor

This set of notes is an attempt to write down what happens in my first-semester college geometry course, Math 341: Modern Geometry I, at the University of Northern Colorado. This course has evolved as I have taught it more than 15 times in the last eight years. The class is directed primarily at pre-service secondary math teachers and pre-service elementary teachers specializing in math. This class has ranged in size from as few as 12 students to as many as 35. It works best with a class of 18–21. The class is an inquiry-based class in which students spend most of their time in class working on problems in groups and then write them up. Because this type of class can be difficult to learn to teach, this set of notes is an attempt to write down as much as possible of what happens, in order to allow other people to try to use the same materials. The class materials are loosely based on those in the book *Experiencing Geometry* [Henderson & Taimiņa, 2005].

In this instructor’s version of these notes, I have included both the materials that I give to the students and also some teaching notes intended for course instructors. A separate student version of these notes contains only the student materials. Sections of these notes that appear in both versions have the same section number in both versions. In this instructor’s version, many of the assignments are followed by teaching notes intended for course instructors. These notes are in blue, and do not appear in the student version. In addition, this instructor’s version also includes this “Introduction for the Instructor,” as well as several extra appendices of materials that instructors may find useful.

It is possible to use individual assignments from these notes to supplement a course that is organized differently. However, I have tried to put the activities together in a way that adheres to the NCTM Curriculum principle, so that they are all interrelated and as a whole tell a coherent story that is larger than the sum of the individual parts. I have tried to articulate some of this larger story in Appendix G. Thus, some of the activities may work better in context of this whole class than they would when used in isolation.

Many of the readings for the course are in the public domain and are included with these notes. However, one of the most important components of the course is readings from the book *Experiencing Geometry In Euclidean, Spherical, and Hyperbolic Spaces* [Henderson & Taimiņa, 2005]. I have included the statements of some of the problems from this book in appendix D. However, the students need to also read the extensive commentary that comes along with these problems in the book. In order to make this possible, you can either require that students buy this book, or else you can arrange for your campus store to obtain the rights to reprint those sections of the book that are needed in a course packet. If you are planning to use these notes as part of a two course sequence using my Modern Geometry II notes, then it makes the most sense to ask the students to buy the book, since it is required for the second course. If you are just teaching a one semester course and not planning to use any other parts of the book, it may make more sense to arrange to print a course packet, since only a small portion of the book is used. It is generally much cheaper for the students if you to obtain the rights to reprint the twenty pages or so that are

needed rather than requiring them to buy the whole book. Arranging for a course packet also allows you to remove part b of Problem 2.1 and the associated commentary, which I feel gives away too much information. When putting together a course packet, I normally include the introductory “Message to the Reader,” Chapters 1, 2, 3, and Problem 7.1. If you are planning to skip the optional assignment on the Vertical Angle theorem given in Section 6, then you can leave out Chapter 3.

Two of the assignments make use of Geometer’s Sketchpad, so you will need to arrange to have class meet in a computer lab with Sketchpad installed on the days you will be working on these assignments. I find that students can generally complete these assignments using Sketchpad in computer labs, but some students prefer to have their own copies. I therefore usually ask the campus store to have the student version of Sketchpad available as an optional course material. If your school doesn’t have a site license for Geometer’s Sketchpad, you could easily adapt the two computer labs to be done with any other dynamic geometry software such as Cabri or GeoGebra. GeoGebra is free and available for download from <http://www.geogebra.org/>.

The following are some of the policies that I put in my syllabus. It is, of course, possible to use some or all of these assignments in a course with significantly different policies.

Writing Assignments: Over the course of the semester, I will be assigning a series of problems for you to write about. Some of these will be informal writing assignments that will be graded only on how complete they are; others will be formal writing assignments, for which you will be expected to turn in a typewritten paper that will be graded not only on the completeness and correctness of your answer, but also on the clarity of your explanations. All of these problems will go through a process of revision: I will make comments on them, and then you will have an opportunity to revise them. You should explore each question and write out your thinking in a way that can be shared with others. Focus on your own ideas and understandings, and turn in whatever your thinking is on a question, even if only to say, “I do not understand such and such” or “I am stuck here.” Be as specific as possible. Conjecture. Use pictures. Respond to my comments and questions. Only the final draft of the formal writing assignments will be graded. There will also be a take home midterm and a final group project that will be presented during our scheduled final exam meeting time.

At the end of the semester, I will ask you to turn in your portfolio containing all the work that you have done for this class, including all drafts of all papers, so please save everything. On average, you should expect to spend at least nine hours per week outside of class on this course. If you are concerned about the time that you are spending on this class, come see me.

Due dates: All assignments will have assigned initial due dates. Revisions of formal writing assignments will generally be due sixteen days after the previous draft. Only the final draft will be graded. Informal assignments can be revised as many times as you like; however, they should be substantially done within sixteen days of the original due date. Progress made after this date will count half as much as progress made before this date, unless the problem is at least $\frac{3}{4}$ complete by then.

Hints for Success: The best approach is to strive for a solid understanding of the course topics and to accept at the start that this necessarily entails some struggling with ideas and feelings of frustration. The course problems take time, especially time to explore and think about the ideas. Often you will need to walk away for a while or for a day, and return to a problem for a second or third look before writing up your response. Expect this. However, do not get behind on the problems. Try to cultivate an approach that is a nice balance between “just getting it done” and avoiding it altogether. Stay connected, and come see me

if you are having difficulties.

Texts: The readings for this course will be excerpts from a variety of sources, including *Experiencing Geometry In Euclidean, Spherical, and Hyperbolic Spaces* by David W. Henderson and Daina Tamina, *Flatland* by Edwin Abbott, and from Euclid's *Elements*.

Course Materials: You will need a folder or binder in which to keep your written work and a disk on which to keep your work done on the computers. You will also want to have a sphere of some kind to look at when we study spherical geometry. Tennis balls work well, as do the plastic spheres often available at craft shops.

Group Work: We will often work in groups in this course. Whenever a group hands in a written assignment, they are required to put on the paper the names of those who participated fully, and only those names. Each person must sign the final copy. Your signature certifies that you participated equally in the project. It is dishonest to turn in work that is not solely and equitably the creation of the team members. You are not required to include on the report the name of someone who started but did not finish, or who did not contribute their share.

Outside Sources: A central aim of this course is to help you learn to develop your own ideas about mathematical questions. You therefore should NEVER consult any reference materials outside of the course texts in answering questions for this course. This includes materials found on the internet. The ideas that you present should be your own.

Further General Teaching Notes

Because this class is non-traditional, some further notes on how the class is run are included here. I encourage new teachers of these materials to consider these ideas, but to find a way to run the class that works for you and your teaching style.

The course is structured around the class assignments. Typically, for each assignment, I start by talking generally about the assignment and putting it into the context of other things we've been doing, and/or with a whole class discussion of some ideas leading into the assignment.

We generally then work on the assignment in groups. In my experience, groups of three work significantly better than groups of two or four. On the one hand, groups of three generate more ideas than groups of two; and on the other hand, it is easy for group members to be left out or to not fully participate in a group of four. I try to change the groups frequently, so that everyone in the class works together at one time or another. Student comments about the class indicate that while students are often hesitant at first about working in groups and about changing groups so often, once they see how productive it can be it ends up being one of their favorite aspects of the class. I have found that the easiest way to quickly form new random groups is by having the class count off (by the number of people in class divided by the number of people per group) and then assign each group number to a specific table or area of the room. The class works best when it meets in a room with tables, with enough tables that each group can have a table to itself.

While the groups are working, I walk from group to group and discuss with them what they are doing. If they are stuck, I ask them to tell me about what they have tried so far, and try to ask questions that will lead them in a productive direction. In any case, I try to know what all the groups are working on at any given point in time. If many groups are stuck in a similar place, I may interrupt the group work to have a whole class discussion. This allows for cross-fertilization of ideas between groups.

The total size of the class makes a big difference in how well the group work goes. A

class of 18 is ideal. As the class size gets larger, it gets harder to get around to all the groups fast enough. One of the goals of the class is to have the students have a positive experience with group work and with discovery learning; in order to achieve this, the instructor needs to be ready to intervene whenever a group is stuck and isn't being productive. If the class is too large, it is harder to do this. I have, however, successfully taught this class to a group as large as 35 students. On the other hand, if the class gets smaller than 18, there is a danger of spending too much time with groups and not allowing them to work on their own. If this starts happening, I will sometimes just leave the classroom for five minutes at a time and go do other things, to give the groups enough time and space to work on their own.

The physical layout of the classroom also makes a big difference in how the class goes. Ideally, the whole course would be taught in a classroom with tables, with enough tables that each group can be by itself at a table. Even better is if the classroom also has computers around the outside for the computer labs and so that groups that are writing up their work together can work on the computers.

In interacting with students in groups and in class discussions, I try to always treat them as the experts and myself as a helpful questioner. So, for example, I always let groups fully explain to me what they are doing as though I have never seen their approach before, even if their approach is one that I have seen many times before. I try not to let them know from my affect whether or not I think their approach is correct until they have explained it to me. This creates an environment in which students take responsibility for explaining their ideas and expect to have them thoughtfully evaluated by themselves, their peers, and the instructor.

After we have spent a reasonable amount of time working on an assignment in class, writing it up will be assigned as homework. Since not all groups will work at the same pace, they won't all be in the same place. Often, the fastest groups will be able to start writing their answers up in class, while the slowest groups may or may not finish answering all the parts of the assignment. Usually I have some minimum core of the assignment in mind, and we don't stop working on it until all the groups have completed this core.

The write-ups are sometimes done in groups, and sometimes individually. Generally, I have more individual write-ups at the beginning of the semester, and more group write-ups later in the semester when the students have more experience with writing in this class and with working productively in groups, and I trust them more.

When informal assignments are submitted, I make a lot of comments on them, and grade them using a "circle" grading method in which each assignment is assigned a grade that is part of a circle. The percentage of the circle that is filled in corresponds to how close the assignment is to being completely correct. I then allow the students to respond to my comments and resubmit their assignments as many times as they would like to. Each time they resubmit an assignment, their grade can improve. The object of this grading system is to get students to focus on fixing their mistakes, and to (only) reward real progress made on an assignment. I also officially have the rule that students must obtain $\frac{3}{4}$ of a circle within two weeks of the first draft being handed back, or else further progress after this point only gets half credit. This rule is there to make sure that students don't leave revisions to be done too long. I don't enforce the rule strictly if a student is making a reasonable effort to get revisions done and just hasn't gotten far enough.

Formal assignments are graded on both correctness and clarity of explanation. I write comments on the first draft, and then grade (but don't write many comments on) the second draft. I often allow students to submit a third draft if they haven't done well on the second draft provided that they come to my office hours to discuss the changes that need to be made first.

For both kinds of assignments, when I get tired of writing the same comments over and over again on student's papers, I compile lists of comments that I frequently make, type them up, and give them to the students, so that I can refer to them on papers just by writing a boxed code. Students don't take the boxed codes quite as seriously as they do handwritten comments, but it saves a lot of writing. I have included such comment sheets for most of the assignments in this course in Appendix A.

Chapter 1

Viewing Tubes

One of the guiding principles of this course is: *Mathematics is not a spectator sport!*

So let's just jump in.

Most of us at one time or another have used a cardboard or plastic tube as a telescope. Although the tube does not actually enlarge what we see, it does help us focus on a narrow field of vision. In this activity, we are going to try to understand and predict how much of a scene is visible through a viewing tube at a distance.

In order to do this activity you will need a measuring stick and several tubes of various dimensions.

1. First, try to identify all the variables that you think might effect how much of a vertical wall is visible through a viewing tube. These may be properties of what you're looking at (*e.g.*, how far away is it), or properties of the particular tube that you're using.
2. Next, pick a tube and gather data with your particular tube. Since you are using one particular tube, some of your variables will be fixed. Your goal is to see what patterns you can observe. With your group you should try to develop a method of making measurements that is as careful and consistent as possible. Make sure that you take enough measurements with your tube so that you will be able to identify patterns. If you only have a couple of measurements, then you won't be able to look for patterns.
3. After you have looked for patterns with an initial tube, try other sizes of viewing tubes and try to see how this changes the patterns that you find. See what conjectures you can make. Our ultimate goal is to be able to predict the field of vision for any viewing tube. That is, if someone shows you a tube and tells you it is a certain distance from the wall, can you find a method of predicting how much of the wall will be visible?

Write up: Explain what you did, what you found, the conjectures that you made, and whatever evidence you have that they are correct. You should try to write your explanation so that it would convince one of your peers that your conclusions are correct.

Teaching Notes: I normally start this activity on the first day of class, after I have gone over the syllabus. It is a good starting activity because it gets everyone up and doing something, and they can start by just taking measurements. It leads eventually to some

nice mathematics that should be accessible to a wide spectrum of students. I try to make sure that every group has taken reasonable data with at least three tubes and has graphed their data. Once they have some data, I try to get them to start making conjectures and then trying to convince themselves they are always true. Some groups may be able to guess a correct formula from their data, while other groups will probably be able to derive a formula from a diagram and then check their data against it. Students sometimes derive the formula using similar triangles, and sometimes using trigonometry (and occasionally other ways, as well).

On the first day of class, I normally spend some time going over the syllabus and course policies, and then jump right into this activity. I tell the students that the word "geometry" means "Earth Measurement," and so we are starting with a measurement activity. I normally give the students some of the second class to work on this as well, and then have each group present what they have done to the class at the beginning of the third class. This usually provides some more ideas to students who are stuck.

At some point, I show the students that a version of this activity is included in the NCTM *Navigations* series [Day et al., 2001] as an example of a best-practices high school geometry lesson.

In order to be considered complete, a student's write up must contain:

- a reasonable description of their method of collecting data;
- a chart of their data that includes a sufficient number of data points;
- a graph of their data;
- a correct formula for computing the viewing diameter;
- a complete proof of why their formula must always work that proceeds from reasonable assumptions that are explicit (*e.g.*, if triangles are claimed to be similar, this must be justified by referring explicitly to the angle-angle triangle similarity criterion); and
- an analysis of how closely their measured values and predicted values agree (for example by computing percentage error for each measurement).

Chapter 2

Formal Assignment: Straightness on the Sphere

Integrate answers to *Experiencing Geometry* Problems 1.1 and 2.1 into one coherent whole that explains what it means for a line to be straight, and uses this definition to explain which lines are straight on the sphere, and why. Your paper should answer all three parts of Problem 1.1. One of these parts is to write a formal definition of what it means for a line to be straight on any surface. You should then use your definition to check if the great circles are straight on the sphere, and if the latitude circles are straight.

Teaching Notes:

Before we do this activity, I have the students read the first six sections of *Flatland* [Abbott, 1885], which is reprinted in Appendix B, and we discuss it briefly. I tell the students that as we are exploring new surfaces this semester, we will want to think about them from the point of view of flatlanders living in the surfaces.

I usually start this activity by putting the students in groups of three and explaining to them that we want to explore geometry on the sphere because it is the geometry of our planet, and because it is a new kind of geometry that very few people in the class will have studied before, so that they can have an experience like the one their future students will have when encountering Euclidean geometry for the first time. I tell them that if we want to study geometry on the sphere, we will have to decide what to use for straight lines, since the objects we study in geometry tend to be made of straight lines. So we must start by trying to decide what lines are straight on the sphere. We quickly realize that to answer this question, we need to decide what the word “straight” means in general. So we consider both of these questions together. I tell my students that working on this problem may feel uncomfortable and weird at first, because probably no one has ever asked them to write a definition or confront a question like this before; I ask them to trust me when I tell them that it is a fruitful question, and one that we absolutely need to confront before we can make progress on the sphere.

I normally let them work on these questions in groups with plastic Lénárt spheres for about 20 minutes the first day, and then send them home to read Problems 1.1 and 2.1 from *Experiencing Geometry* [Henderson & Taimiņa, 2005], which give them a lot of new ways of thinking of these questions (but very few answers). The statements of these Problems are included in Appendix D. However, I ask my students to read not only the statements of the Problems, but also the extensive commentary on them included in the book. If I am making a course packet that only includes excerpts from the book, I take out the part b of Problem 2.1 that tells the reader that the lesser (latitude) circles are not straight. I prefer to leave this as a question until the class as a whole has reached a consensus about which

lines are straight. At some appropriate point while they are working in their groups on the first day, I introduce the terms *intrinsic* (from the point of view of a flatlander living in the surface), *extrinsic* (from the point of view of someone looking at the surface from outside), *great circle* (a circle on the sphere whose extrinsic radius is the same as that of the sphere), and *lesser circle* (any circle whose extrinsic radius is less than that of the sphere). I try to introduce these terms when it seems like the underlying ideas have started coming up in the group discussions. I point out that any definition of what is straight on the sphere will have to be an intrinsic definition, since there are no lines that are extrinsically straight on the sphere.

After they have done the reading from *Experiencing Geometry*, I hand out the official statement of this assignment, and then we generally spend one to two more days working on parts of it in class. When I think that the class needs more specific direction, I ask them to focus on one of the particular pieces of Problem 1.1. I might give them twenty minutes in their groups to come up with all of the symmetries of a straight line that they can, and then discuss them as a class. Or I might ask them to spend time in their groups coming up with ways to practically create straight lines or check that a given line is straight. Or we might spend time discussing what makes a good mathematical definition, and compare possible definitions of some other idea in geometry. For example, we might try to come up with possible definitions of a square and compare them. I try to emphasize the idea that a mathematical definition, unlike a dictionary definition, should be checkable in some unambiguous way.

When I think that all of the groups have gotten far enough in their thinking about these questions, we move on to the next activity, and they work on writing up their papers on their own. Even though we work on this in groups in class, I prefer to have the papers written individually. In this way, each student is held accountable for understanding what their group did. In general, I start the semester with several individual assignments, but move towards more group assignments as the semester progress and as my trust in the students' abilities to work productively in groups grows. Because this is a formal assignment, I make comments on the first draft (but don't grade it); and then grade the second draft (but usually don't make many comments on it). My intention is to separate the thinking about how a paper could be improved from the final, summative evaluation of the paper. Although I originally tell students that this formal assignment will only be turned in twice, after I give the graded second drafts back, I tell the students about my grading rubric, and tell them that if they would like to revise the papers one more time, they can if they first come to my office hours to discuss exactly what needs to be fixed. Sometimes I only extend this offer to people whose original grade is below a certain threshold.

See Appendix A for the sorts of comments I typically make on students' papers.

My grading rubric is as follows: in order to be deemed satisfactory, a paper must:

- address all three parts of Problem 1.1 in some reasonable way;
- propose a reasonable definition for straightness that is at least mostly checkable; and
- correctly use this definition to check that the great circles on the sphere are straight, and the lesser circles are not.

A paper that minimally meets these criteria gets a B. A papers gets a lower grade if it doesn't meet these criteria, and gets a higher grade if it meets them in way that is thoughtful, clear, elegant, and/or well explained.

2.1 Euclid's Elements

Before moving on to the next assignment, we next spend some time looking at Euclid's *Elements*. We generally use Sir Thomas Heath's relatively literal translation [Euclid, 1956], the relevant parts of which are reproduced in Appendix C. Typically, I have the students initially read the definitions, postulates, common notions, and Proposition 1. We discuss these in class and then transition to using Geometer's Sketchpad. After the students have completed the first Sketchpad tutorial (which comes with Sketchpad and involves constructing a square), I ask the students to use what they have read in Proposition 1 to construct an equilateral triangle in Sketchpad. This makes a natural transition to the first computer lab discussed in the next section.

Later in the semester, we return to the Elements and discuss Proposition 4, which is about Side-Angle-Side triangle congruence, and Proposition 5, the Isosceles Triangle Theorem.

Euclid's *Elements* consists of twelve books. The first four books are about elementary planar geometry; books five through ten are largely about the theory of ratio and proportion and about what would now be considered number theory; and books eleven through thirteen are about three dimensional geometry, and work their way up to the constructions of the five platonic solids.

Euclid starts by giving twenty-three definitions, along with five postulates and five common notions. The definitions, postulates, common notions, and propositions 1, 4, and 5 are given in Appendix C.

In discussing Euclid's definitions, I try to lead the students to noticing that they are really of two types. Some of his definitions, such as that of a circle in Definition 15, give specific defining properties that the object defined must have. These definitions are concrete and are sometimes used in his proofs, and are like the mathematical, checkable definitions that we asked the students to write for the Straightness on the Sphere assignment. Other definitions, like that of a point given in Definition 1, give more general descriptions of the objects being described. They describe an object, but not in terms of properties that could be concretely used in a proof. A reader who already knows what a point is will probably agree that it "has no part," but a reader who didn't probably wouldn't find this definition very useful. These are more like dictionary definitions that describe an object in a way that is not unambiguously checkable.

Euclid's Postulates and Common Notions set down the assumptions that he will use in his proofs. The Common Notions set down assumptions that are supposed to be true in general, while the Postulates set down the assumptions that are supposed to be particular to geometry. Thus, for example, Common Notion 1 states the transitive law of equality, which is true throughout mathematics.

The first three postulates are construction postulates, that state allowed constructions that could be done with a straight edge and compass.

Postulate four may seem trivial to many students on a first reading, but it really is not. Euclid's definition of a "right angle" is not one that is ninety degrees, but rather one that divides a straight line into two equal pieces. Thus, the postulate says that all such right angles, and therefore all straight lines, are the same everywhere. In effect, it says that the surface is uniform; later in the course, we see that it fails on surfaces like cones where some points are different than others.

The fifth postulate is considerably more complicated than the others, and can be very difficult for students to parse. This is unsurprising, since, historically, many people viewed this postulate as being too complicated to be taken as an assumption, and tried to prove

it from the other four postulates. The discovery of hyperbolic geometry in the nineteenth century showed that it cannot be derived from the other postulates, however, since the first four postulates are true in the hyperbolic plane, but the fifth one isn't. The fifth postulate is an if-then statement, and therefore gives students an opportunity to grapple with what makes an if-then statement true or false. A very common mistake is to try to use the postulate in cases where the interior angles do NOT add up to less than two right angles.

Chapter 3

Computer Lab I: Symmetries of Triangles and Quadrilaterals

In this lab we will be examining the symmetries of different kinds of triangles and quadrilaterals.

1. Construct an equilateral triangle that stays equilateral no matter how you move the points around. (To do this, start with a segment and use the construction that Euclid describes in his Prop. 1.) Likewise, construct a square that remains a square. (You'll need to think about how to do this. You'll want to start with another segment, and construct a square on the segment. One useful command for this is "Construct Perpendicular Line," which is in the construct menu.) Save your sketch as sym1.

2. For each of these, see how many symmetries you can find. Construct a new arbitrary line, and reflect the figures over it. To do this, select the line, then go to "Mark Mirror" in the Transform menu. Then, select the things you want to reflect, go to the Transform menu, and choose "reflect." After you have reflected the figures over the line, you can move the line around, and see what reflection symmetries you can find. When you find one, the figure will coincide with its reflection. Also, create a new point and a new angle (made from two line segments), and rotate the figures about the point by the angle, using the transform menu. Again, see if you can find symmetries by making the figure coincide with its rotation.

3. Now go back to your saved sketch sym1. For each symmetry that you found, construct the line/point of symmetry directly from the original figure, and reflect/rotate the figure about that line/point. If you've constructed it correctly, the image should stay on top of the original figure no matter how you move it. This will give you a precise way to specify the line/point of symmetry by saying how to construct it.

4. Finally, for each symmetry that you found, try to see what kinds of triangles or quadrilaterals have that symmetry. You can do this by starting with a line or point of symmetry and a point or segment of a figure, reflecting/rotating it about the point or line, and seeing how much of the figure is determined. What happens if you require two or more symmetries?

Homework: Write up a lab report explaining your results. You can do this by yourself or in pairs. Your report should include: 1) A description of each of the symmetries that you found, including explicit instructions for how to find/construct the points, lines, and angles of symmetry; and 2) For each set of symmetries on your list, a complete list of all of the types of triangles and quadrilaterals that have that symmetry.

Teaching Notes:

The first half of this assignment is designed to force students to wrestle with their definition of what a symmetry is, and to see how concrete this definition is. Part four of the assignment is much more difficult, as it asks the students to go from the symmetries to the objects. Students' first inclination is always to try different shapes to see what symmetries they have, which leads to a discussion of whether or not this will allow us to ever be sure that we have found all such shapes. However, students should be able to eventually generate all the shapes by starting with (for example) a line of symmetry and then putting down points on and/or off of the line of symmetry, reflecting them over the line, and then connecting them to form a shape that has the line as a line of symmetry. Moving the points to all possible positions gives all of the possible shapes with that line of symmetry. Most students require a fair amount of prompting to find this approach.

There are also some nice connections to Abstract Algebra that can be discussed here. The shapes that have some but not all of the symmetries correspond to subgroups of the group of symmetries of the square (or equilateral triangle).

Students normally work on this assignment in pairs at the computers, and also turn it in in pairs.

Chapter 4

Euclid's Postulates on the Sphere

Informal Assignment: Which of Euclid's Five Postulates are true on the Sphere? Also look at the following postulate:

Postulate 1'. Two points determine a unique line.

Teaching Notes: This assignment combines our work with Euclid's postulates and our work on the sphere. Postulate 1 is true, but students have a hard time giving a justification for why. I try to steer them towards giving a concrete method of connecting any two given points. Postulate 1' is false. If we have two non-antipodal points, they can be connected a short way and a long way. If students come up with this example, I tell them that those are different line segments, but not different lines, so this isn't a counterexample to the postulate. The antipodal points provide the counterexample. Postulate two is true because although straight lines have finite total length, it is still true that any line segment can be extended indefinitely in either direction (with overlap if necessary). When looking at postulate 3, students are often surprised to notice that on a sphere circles can also be straight lines or individual points. Postulate four is true, and I make students justify this by going back to Euclid's definition. Postulate five is true because all lines intersect on the sphere.

Chapter 5

NG worksheets

The worksheets in this section were written by Charles Coppin, and are used here with his permission.

5.1 Logic Worksheet

An axiom system has two components:

- A set of special words which we officially assume have no meaning. The words are sometimes called “undefined terms,” “primitives” or “technical terms.”
- A set of statements which we assume to be true concerning the “undefined terms.”

One of our favorite examples of an axiom system is as follows:

Neutral Geometry Axiom System (NGAS). Suppose “point” and “line” are undefined. Lines will consist of points. Consider the following axioms:

Axiom 1. If L is a line, then there exists at least two points belonging to L .

Axiom 2. If L is a line, then there exists at least one point which is not on L .

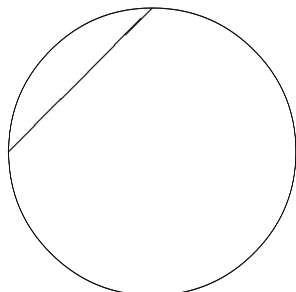
Axiom 3. There exists at least one line.

Axiom 4. If A and B are different points, then there exists at least one line which contains both A and B .

An **interpretation** of an axiom system is a set of meanings for the undefined terms. These meanings are assigned without any regard for whether the axioms become true statements about the undefined terms. A **model** for an axiom system is an interpretation for the axiom system in which the axioms are true statements about the undefined terms.

Problem 1. By “point” we mean Calvin, Hobbes, or 1; that is, the set of points is $\{\text{Calvin}, \text{Hobbes}, 1\}$. By “line”, we mean one of the following combinations: $\{\text{Calvin}, 1\}$, $\{\text{Hobbes}, 1\}$. Thus, we have given meaning to “point” and “line.” Do we have a model for NGAS? If not, explain which axioms fail to be satisfied.

Problem 2. Consider the circle below:



By “point” we mean a point on the circle or interior to the circle. By “line,” we mean a chord of the circle. Do we have a model for NGAS? If not, explain which axioms fail to be satisfied.

Problem 3. Consider a fictitious goldfish bowl containing two goldfish in water and sitting on a table. Suppose there is one goldfish on the table beside the bowl. By “point” we mean one of the goldfish and by “line” we mean the two goldfish in the bowl. Do we have a model for NGAS? If not, explain which axioms fail to be satisfied.

5.2 NG Worksheet 1

Neutral Geometry (sometimes called Absolute Geometry) is that geometry which is common to both Euclidean and Non-Euclidean geometry. We start with some familiar axioms. Respond to the problems below.

“Point” and “line” are undefined. Lines will consist of points.

Axiom 1. If L is a line, then there exist at least two points belonging to L .

Axiom 2. If L is a line, then there exists at least one point that is not on L .

Axiom 3. There exists at least one line.

Problem 1. What is the minimum number of points possible based on the Axioms 1, 2 and 3 above? Prove your answer.

Problem 2. State the theorem you discovered and proved in Problem 1.

Problem 3. Find a model which satisfies Axioms 1 and 3 but not Axiom 2.

a. “point” means ...

b. “line” means ...

Explanation:

5.3 NG Worksheet 2

The results of NG Worksheet can be applied to this one. That is, the "truths" of that worksheet can be used in arguments here.

Problem 1. Create a model for Axioms 1, 2, and 3 in which there are at least two points x and y with the property that no line contains both x and y .

"point" means ...

"line" means ...

Explanation:

Problem 2. Is the following a theorem of the axiom system?

Each point belongs to some line.

Prove your response.

5.4 NG Worksheet 3

I think you would agree that any geometry which contains two points not belonging to any line is not very interesting. We have an unacceptable situation as represented by Problem 2 of NG Worksheet 2. Therefore, we need the next axiom.

Axiom 4. If A and B are different points, then there exists at least one line which contains both A and B.

Problem 1. Explain what is wrong with the following proof that each point belongs to some line. Find the invalid line of the proof.

By Axiom 3, there is some line L. By Axiom 1, L contains two points A and B. Thus, for any point A, there is a line containing A. Q.E.D.

Explanation:

Problem 2. State the theorem that we attempted to prove in Problem 1. Now, prove it.
Statement.

Proof.

5.5 NG Worksheet 4

Problem 1. In Problem 2 of NG Worksheet 3, we proved that each point belongs to some line. Which of the following is true of the axiom system?

A. There is a model of the axiom system in which at least one point belongs to only one line.

B. It is a theorem that each point belongs to at least two different lines.

Prove your response.

5.6 NG Worksheet 5

Problem 1. Is it a theorem of the axiom system that each point belongs to at least three different lines?

Prove your response.

Problem 2. Find a model which satisfies the axiom system in which there are two different points A and B such that at least two different lines contain both A and B.

5.7 NG Worksheet 6

Summary of NG Worksheets 1 – 5:

“Point” and “line” are undefined. Lines will consist of points.
Axiom 1. If L is a line, then there exist at least two points belonging to L .
Axiom 2. If L is a line, then there exists at least one point that is not on L .
Axiom 3. There exists at least one line.
Theorem 1. There exist at least three different points.
Theorem 2. If P is a point, then there is a line containing P .
Theorem 3. If P is a point, then there exist two different lines containing P .
Axiom 4'. If A and B are different points, then there exists one and only one line which contains both A and B .

Notice that we have replaced Axiom 4 with a new axiom, Axiom 4'.

Problem 1. Give a definition of what we would mean by the statement that a set of points is not collinear.

Problem 2. Are there three non-collinear points? Prove your answer.

Problem 3. Axiom 2 says that there is always some point not on a given line. Can we turn that statement around and say that there is a line not containing a given point? Prove your answer.

5.8 NG Worksheet 7

Problem 1. Suppose L and H are lines. What can you say about the maximum number of points, which can belong to both L and H ? Prove your answer.

Problem 2. Find a model with the smallest number of points, which satisfies Axioms 1, 2, 3, 4'. Prove your response.

Teaching Notes:

These neutral geometry worksheets, written by Charles Coppin, are taken from a course taught at the University of Dallas. This course is unusual in that it is the main course that meets their gen-ed math requirement.

I normally have students work on these in groups, and every so often I interrupt them and have student volunteers go to the board to present problems. Because we are using such a simplified axiom system here, it is a great place to talk about writing proofs, the role of axioms in proofs, and the relationship between axioms, proofs, and models. I tell the students that when they have completed the worksheets, I will then tell them which problems I will collect and grade. Most often, I collect and grade NG worksheet 4 and NG worksheet 6, problem 3.

Chapter 6

Formal Assignment 2: Vertical Angle Theorem

For this assignment, you should define what you think an angle is, and use your definition to prove the Vertical Angle Theorem.

In doing this, you can use Problems 3.1 and 3.2 from *Experiencing Geometry* as a guide, but you need to integrate your answers to the various questions into a unified, coherent whole. You should try to make your proof as clear and convincing as possible. The best proofs are those that not only convince the readers that something is true, but also allow them to understand why it is true. You should bear this in mind when you're deciding what assumptions to make and what kind of a proof to give. This is one reason that you want your assumptions to be simpler than the fact that you're trying to prove, and as obvious as possible. (It's also one reason that different people will give different proofs if they think of angles in different ways. Explaining why two angles have the same shape is very different than explaining why they have the same degree measure.) If you do decide to give a proof using degree measure, be careful not to confuse an angle (which is some kind of geometric object) with its degree measure (which is a number). If you want to talk about measuring angles, you'll need to explain how you want to assign a measurement to each angle, and also why angle measurements under your definition have any properties that that you need for your proof. This is one reason that the authors suggest that proofs using symmetries are "generally simpler" than proofs involving measuring angles.

Teaching Notes:

The statements of the two Problems from [Henderson & Taimiņa, 2005] that are mentioned in this problem are reproduced in Appendix D.

This assignment is one that I often omit or replace with the next one, depending on my class and how the semester is working out timewise. When I do assign it, I have the students read *Experiencing Geometry* Problems 3.1 and 3.2. We talk about the VAT in class, and I have someone share the typical high-school proof of the VAT. ($A + B = 180^\circ = B + C$ so $A = C$.) I then give them this assignment, tell them that they have to find a proof that is different from the typical high-school proof, and that we aren't going to discuss this any further until the first draft is due, usually two weeks later. On the day the first draft is due, I have them pair up and read and comment on each other's papers, using the form in the next subsection. I then have as many people as possible present their proofs at the board and we discuss them. At the end of the class, I tell them that anyone who wants to can revise their paper once before turning in the first draft at the following class. If I omit this assignment and do the next one instead, I do a similar peer review activity for that assignment.

My grading rubric is as follows: in order to be deemed satisfactory, a paper must:

- include a checkable definition of what constitutes an angle;
- include a checkable definition of what it means for two such angles to be congruent (and this definition must be consistent with their definition of an angle); and
- use these definitions to make a reasonable argument that any two vertical angles must in fact be congruent.

A paper that minimally meets these criteria gets a B. A paper gets a lower grade if it doesn't meet these criteria, and gets a higher grade if it meets them in a way that is thoughtful, clear, elegant, and/or well explained.

6.1 Vertical Angle Theorem Peer review questions

Your name:

Reviewing a proof by:

According to this paper, an angle is:

According to this paper, two such angles are congruent if:

Sketch the author's proof here:

Is the proof general—that is, does it work for any two vertical angles?

If the proof is an algebraic proof using measurements of angles, has the author explicitly told you how to find the measurement of a given angle?

If so, explain his or her method here:

What assumptions does the proof rely on? Does the author make these assumptions explicit?

Chapter 7

Formal Assignment 3: Side Angle Side on the Sphere

Recall the following fact about planar triangles:

(SAS) If two triangles have the two sides equal to two sides respectively, and have the angles contained by the equal straight lines equal, they will also have the base equal to the base, the triangle will be equal to the triangle, and the remaining angles will be equal to the remaining angles respectively, namely those which the equal sides subtend.

This is Euclid's Proposition 4.

Decide whether or not SAS is true on the sphere. If you think that it is true on the sphere, then give a convincing argument that shows that it is true. If you think that it isn't true, explain why it is false, show how to modify the statement to make it true on the sphere, and then prove your modified statement. In either case, explain and justify any decisions that you had to make in order to come to your conclusion. Try to write explanations that would convince a reasonable skeptic. You will probably find it helpful to thoroughly understand Euclid's planar proof before trying to decide what happens on the sphere.

Teaching Notes:

This is a nice assignment because it makes students look at the relationship between proofs, models and counterexamples. Hopefully, they eventually see that because there are counterexamples to SAS on the sphere, Euclid's proof can't work as written on the sphere, and that a modification that eliminates the counterexamples should also make it possible fix the proof so it does work on the sphere. We often have nice discussions about what it means to different people to look for the "best" way to fix the theorem to make it true on the sphere.

When I do the peer review activity with this assignment, I use the modified questions given in the next subsection.

My grading rubric is as follows: in order to be deemed satisfactory, a paper must:

- show at least one correct counterexample to Side-Angle-Side on the Sphere, and explain why it is a counterexample;
- explicitly propose a fixed version of Side-Angle-Side on the Sphere or else explicitly

redefine what a triangle is in a way that makes Side-Angle-Side true on the Sphere;
and

- give a correct proof of their corrected theorem.

A paper that minimally meets these criteria gets a B+. (These criteria are harder to meet than the criteria given for the other formal assignments.) A paper gets a lower grade if it doesn't meet these criteria, and gets a higher grade if it meets them in a way that is thoughtful, clear, elegant, and/or well explained.

7.1 Side Angle Side Peer review questions

Your name:

Reviewing a proof by:

According to this paper, a triangle is:

According to this paper, is SAS true or false on the sphere?

What justification has the author given for this conclusion? Do you find it convincing?

If the author concluded that SAS was false, have they explicitly stated a new version of SAS that is true on the sphere? If so, what is their changed version?

If the author concluded that SAS was true on the sphere, have they given a proof for this conclusion? If the author concluded that SAS was false on the sphere but gave a fixed version of SAS on the sphere, have they proven their fixed version? If so, explain their proof here:

If they gave a proof, does it convince you to a strong degree of certainty that no further counterexamples will be found? Why or why not?

Chapter 8

Euclid's Postulates on the 450° cone

Informal Assignment: Which of Euclid's Five Postulates are true on the 450° cone? Also look at Postulate 1'.

Teaching Notes: The 450° cone can be made by taping a quarter of a piece of paper into a slit cut in another piece of paper. See [Henderson & Taimiņa, 2005, Chapter 4] for more details.

Students have to start this assignment by again trying to figure out what the straight lines are on this surface, which is actually one of the hardest parts of the assignment. I remind them that they all wrote general definitions of straightness for exactly this kind of situation, in which we want to move our study of geometry to a new surface, and that they should go back and look at their definitions again. The main question that they have to confront is about what happens to straight lines that hit the cone point; the answer is that in order to have reflection across themselves and half-turn rotation symmetry, the lines must make 225° angles with themselves at the cone point.

The hardest postulate to figure out here is postulate one. If students are having trouble with it, I encourage them to draw lots of points on their cones and try to connect them all. This helps them realize that for any point P other than the cone point, there is a 90° region that can't be reached from P .

Chapter 9

The Area of a Triangle on a sphere

Informal Assignment: Find a formula for the area of a triangle on the sphere. Write up a proof that your formula is correct. Use *Experiencing Geometry* Problem 7.1 as a guide.

Teaching Notes:

The statement of *Experiencing Geometry* Problem 7.1 is included in Appendix D.

I often lead into this assignment with a class discussion about what is the possible range for the sum of the interior angles of a triangle on the sphere. Students can usually conjecture the correct range (the sum must be more than 180° and less than 900°); this range can then be justified by looking at the formula for the area of a triangle on the sphere once they've found it.

This assignment usually ends up being relatively straightforward. It is vital, however, to have the students working on physical spheres and shading in the lunes. When I first started doing this activity, I was always surprised at how hard it is for students to come up with and to justify the formula for the area of a lune on the sphere. Often, the weakest students get the most out of this assignment, in terms of understanding where formulas come from, how they are used, and in terms of using algebra skills in a meaningful context to get new knowledge about an otherwise difficult question.

Once students have found a formula, rather than telling them if it is right or wrong, I have them check it against examples themselves. A good starting example is an equilateral triangle with three ninety degree angles, which takes up one eighth of the area of the sphere.

Chapter 10

Computer Lab 2: Billiards Tables

In this computer lab, we will be modeling the path taken by a ball on a billiards table.

1. Build a billiards table! It should have sides that always stay horizontal and vertical, and you should be able to change its dimensions by dragging the corners. How can you do this? Start with a vertical line and two points not on the line. Construct a rectangle whose upper-left and lower-right corners are the two points and whose sides are all parallel or perpendicular to the vertical line. Save your table as “table1.”
2. Add a point to represent a ball, and another point to show the direction that the ball is being hit in. We want to construct the path that the ball would follow when hit towards the second point. Show the path of the ball until it hits the side of the table. Add the path that the ball takes after its first bounce. Add the path for the next several bounces. Save your table as “table2.”
3. Go back to table 1. Add two balls. Construct the path that one ball has to travel to hit the other directly. Make a button to show/hide this path. (You can construct a button to hide and show objects in Geometer’s Sketchpad by selecting the objects you would like the button to control, and then picking the “Hide/Show” command from the “Action buttons” submenu of the Edit menu.) Construct a path from one ball to the other that bounces once. Construct a path that bounces twice. How many paths are there from one ball to the other that include one bounce? Two? Three? This is actually a practical question: it is common in pool games to want to bank a shot off of a rail—that is, to hit a ball by bouncing the cue ball off of a side of the table. How could a pool player use your methods to find the spot that he or she should aim for in order to hit the ball after one bounce?

Homework: Write a report explaining

- how to construct the path a ball takes after it hits a rail;
- how you know that your method will result in the angle of incidence of the ball equaling the angle of reflection of the ball;
- how to construct a path that connects two given balls with one, two, or three bounces; and
- how many ways there are to connect the balls by a path of n bounces, and how you know.

Teaching Notes: For information about this assignment, please see my article “Visualization on Cones and Pool Tables using Geometer’s Sketchpad” [Miller, 2006], reprinted in Appendix E.

The ideas explored in this assignment can also be found in some secondary curricula, such as the “Reflect on This” lesson in the tenth grade *SIMMS Integrated Mathematics* curriculum [SIMMS, 2003].

Chapter 11

NCTM Standards Assignment

On the next several pages are listed the NCTM Standards for School Mathematics, published by the National Council of Teachers of Mathematics [NCTM, 2000]. This assignment is intended to help you become more familiar with these standards. Start by reading the short general overview of these standards found at

<http://www.nctm.org/standards/standards.htm>.

In order to get access to the full text of the NCTM Principles and Standards document, go to <http://standards.nctm.org/>, and sign yourself up for the 90 day free access.

Once you have access, start by reading the more specific information about the Geometry Standard for whatever grade level you are planning to teach. (To get there, sign in, then click on the appropriate grade level at the top of the page, and on Geometry on the left of the page. If you are not planning to become a teacher, pick any grade level to look at.)

Next, pick three standards that you feel have been addressed in this course. At least one of these must be one of the 4 Geometry Content Standards, and at least one must be one of the 5 Process Standards. (For the Content Standards, we are looking at the individual bullet points; for the Process Standards, we are looking at the whole standard, which contains several bullet points.)

For each standard that you picked, read the full text on the standards website discussing that standard at the grade level you plan to teach, then write two paragraphs about it. The first paragraph should describe how the standard has been addressed in this class, and the second should address some specific ways in which you can address the standard at the grade level you are planning to teach.

11.1 NCTM Standards

Content Standards

Number and Operations Instructional programs from prekindergarten through grade 12 should enable all students to—

- understand numbers, ways of representing numbers, relationships among numbers, and number systems;
- understand meanings of operations and how they relate to one another;
- compute fluently and make reasonable estimates.

Algebra Instructional programs from prekindergarten through grade 12 should enable all students to—

- understand patterns, relations, and functions;
- represent and analyze mathematical situations and structures using algebraic symbols;
- use mathematical models to represent and understand quantitative relationships;
- analyze change in various contexts.

Geometry Instructional programs from prekindergarten through grade 12 should enable all students to—

- analyze characteristics and properties of two- and three-dimensional geometric shapes and develop mathematical arguments about geometric relationships;
- specify locations and describe spatial relationships using coordinate geometry and other representational systems;
- apply transformations and use symmetry to analyze mathematical situations;
- use visualization, spatial reasoning, and geometric modeling to solve problems.

Measurement Instructional programs from prekindergarten through grade 12 should enable all students to—

- understand measurable attributes of objects and the units, systems, and processes of measurement;
- apply appropriate techniques, tools, and formulas to determine measurements.

Data Analysis and Probability Instructional programs from prekindergarten through grade 12 should enable all students to—

- formulate questions that can be addressed with data and collect, organize, and display relevant data to answer them;
- select and use appropriate statistical methods to analyze data;
- develop and evaluate inferences and predictions that are based on data;
- understand and apply basic concepts of probability.

Process Standards

Problem Solving Instructional programs from prekindergarten through grade 12 should enable all students to—

- build new mathematical knowledge through problem solving;
- solve problems that arise in mathematics and in other contexts;
- apply and adapt a variety of appropriate strategies to solve problems;
- monitor and reflect on the process of mathematical problem solving.

Reasoning and Proof Instructional programs from prekindergarten through grade 12 should enable all students to—

- recognize reasoning and proof as fundamental aspects of mathematics;
- make and investigate mathematical conjectures;
- develop and evaluate mathematical arguments and proofs;
- select and use various types of reasoning and methods of proof.

Communication Instructional programs from prekindergarten through grade 12 should enable all students to—

- organize and consolidate their mathematical thinking through communication;
- communicate their mathematical thinking coherently and clearly to peers, teachers, and others;
- analyze and evaluate the mathematical thinking and strategies of others;
- use the language of mathematics to express mathematical ideas precisely.

Connections Instructional programs from prekindergarten through grade 12 should enable all students to—

- recognize and use connections among mathematical ideas;
- understand how mathematical ideas interconnect and build on one another to produce a coherent whole;
- recognize and apply mathematics in contexts outside of mathematics.

Representation Instructional programs from prekindergarten through grade 12 should enable all students to—

- create and use representations to organize, record, and communicate mathematical ideas;
- select, apply, and translate among mathematical representations to solve problems;
- use representations to model and interpret physical, social, and mathematical phenomena.

Teaching Notes:

Students work on this assignment at home, and we generally don't talk about it much until the day they turn it in. Students are generally quite good at identifying how we have addressed the standards in our class, but are not nearly as good at specifying how they would use them in their own classes in realistic ways.

Chapter 12

The Shape of Space

Informal Assignment: Imagine that you are a two-dimensional flatlander living on a two dimensional surface. Suppose that you knew somehow that your universe was either a Euclidean Plane, a sphere, a flat torus, or a 90 degree cone. Write a flow chart showing how you could determine the shape of your universe.

Teaching Notes:

This is a great wrap up assignment for the class, and we usually do it during the last week of class. I tell the students that they don't need to be realistic, and it's fine do things like sending out an infinite number of explorers in all directions, *etc.* I alter the assignment depending exactly which surfaces we have studied in a given semester.

Before we work on this, I usually show the two short videos that come with the *Exploring the Shape of Space* middle-school curricular materials from Key Curriculum Press [Weeks, 2001].

Chapter 13

Final Project: Geometry on the Cone

See how much geometry you can develop on cones with cone angles that are less than 360 degrees (*e.g.*, the 90 degree cone). You may also be interested in comparing these with cones with cone angles that are greater than 360 degrees (*e.g.*, the 450 degree cone) and/or cylinders.

For any of these surfaces you can ask any of the following questions:

- What do the intrinsically straight lines look like? What happens to lines that run into the cone point?
- What do circles look like?
- Which of Euclid's Postulates are true?
- What is the sum of the interior angles of a triangle?
- What is the holonomy of a triangle?
- Which triangle congruence theorems hold (or don't hold)?
- Is there a unique straight line joining any two points? If not, for which points is there a unique straight line joining them?
- Do any straight lines intersect themselves? If so, how many times?

Explore some of these questions, or others, that interest you, and write a paper explaining what you find. You do not need to decide in advance all of the questions you are going to explore; most people find that it is most productive to start with a question that seems interesting, and then see where that question takes you and what other questions it raises.

Teaching Notes:

This assignment is the one we spend the most time on in this class, and I always find it to be the most valuable, as well. I generally allow three to four weeks for this one assignment. It's totally worth it! For lots more information about this assignment, please see my article [Miller, 2006], reprinted in Appendix E.

This assignment can be done without knowing anything about holonomy. When I include the question about holonomy, we spend twenty minutes at the beginning of class one day exploring holonomy on the sphere. For more about holonomy, see Chapter 7 of *Experiencing Geometry* [Henderson & Taimiņa, 2005].

The last meeting of my class is typically during finals week, when, in lieu of an in-class final, groups present their work on this assignment and then I collect their final portfolios. I normally compute final grades as a weighted average of the formal and informal assignment averages, with formal assignments making up 65% of the final grade and informal assignments making up the other 35%. The formal assignments are more heavily weighted since many students are able to get very high grades on the informal assignments since they can keep working on them until they are complete.

Appendix A

Comment Sheets

The following sections contain comments that are frequently made on papers in this course. If you get a paper back with a comment that is just a letter and number in a box, look here to find the corresponding comment.

A.1 Viewing Tubes Comments

V1. Great! Can you graph your data?

V2. How closely do your predicted values match your measured values? Can you compute the percentage error for each of your measurements?

V3. Good! Now can you find a (deductive) explanation for why this formula works? Otherwise, we won't be sure if it always works. A good starting point would be to draw some diagrams of the situation that you're modeling mathematically.

V4. Ok, good. How do you know mathematically, though, that the two triangles are similar?

V5. How do you know that these ratios are equal? What fact about similar triangles are you using?

V6. Are you measuring from your eye, or from the end of the tube?

A.2 Straightness on the Sphere Comments

A1. Good! What property of straight lines is this related to?

A2. Formal assignments should include an introductory and concluding paragraph.

A3. Can you explain the meaning of each of the symmetries you mention?

A4. What does this mean? I'm not sure exactly how to check this.

A5. You've given a general definition of what it means for a line to be straight, and explained which lines are straight on the sphere. However, you still need to show why the greater circles fit your definition, and why the lesser circles do not.

A6. How does this fit with our example of a straight line segment on the sphere that isn't the shortest distance between its endpoints?

- A7. Can you show how great circles have each of these symmetries/properties?
- A8. Can you show how great circles fit each clause of your definition?
- A9. Why aren't lesser circles straight according to your definition?
- A10. How does this relate to your definition?
- A11. Is this your official definition of what it means to be straight? If so, can you make this more explicit?
- A12. I'm still not exactly sure what your official definition of straight is. Can you give an explicit definition? Without this explicit definition, it isn't clear what you need to check to show that the great circles are straight, and the lesser circles are not.
- A13. Can you include (more) discussion of different ways to create straight lines and to check that a line is straight? Also, you should discuss why they work and what properties of straight lines they are related to.
- A14. How would you check this practically?
- A15. You need to address the part of Problem 1.1 that asks you to find all of the symmetries of straight lines.

A.3 Computer Lab I (Symmetries of Triangles and Quadrilaterals) Comments

- B1. You've listed several different kinds of triangles and quadrilaterals and described their symmetries. The question, however, asked something slightly harder to answer: What are all of the triangles and quadrilaterals with one of the symmetries that we're discussing, and can you give a convincing argument that you've found them all? So, to approach this question, you should start with a symmetry, find all of the triangles or quadrilaterals with that symmetry, and explain how you know your list is complete.
- B2. Can you specify exactly what the line (or point) of symmetry is, by explaining how to construct it?
- B3. Bilateral symmetry really just means symmetry across some line. So, can you specify more precisely exactly what lines are lines of symmetry of your figure? Tell me how to construct the lines of symmetry from the figure.
- B4. About what point? Tell me how to construct the point from the figure.
- B5. I don't think your list is complete.
- B6. I don't think you can create all possible trapezoids this way. Which ones can you create?
- B7. I don't understand this.
- B8. So what does this show?
- B9. I don't think you've found all of the triangles and quadrilaterals that have one of these symmetries yet.
- B10. Do all trapezoids have this symmetry, or just some of them?
- B11. It isn't a reflection symmetry of the figure if the reflected image doesn't land directly on the original figure. If you have to reflect it and then translate it back, it isn't a reflection symmetry.

A.4 Euclid's Postulates on the Sphere Comments

C1. Ok—but how do you know that a straight line can be drawn connecting any two points? Can you describe how you would actually do it given two points on a physical sphere?

C2. So is this postulate true or false?

C3. Euclid's definition of a right angle doesn't say that it is 90 degrees.

C4. You still need to argue that all right angles, at all points on the sphere, are equal to one another. You can do this by showing that all angles that meet Euclid's definition of being right are 90 degrees, and therefore they're all equal.

C5. Your example shows that there are two different line segments connecting the two points, but still only one line (great circle). Can you find a different example where two points can be connect by two completely different great circles?

C6. How do you know all right angles are 90 degrees on the sphere? You need to show this using Euclid's definition.

C7. This isn't true on the sphere.

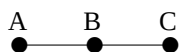
A.5 NG worksheets comments

D1. You're trying to prove a fact about all points. So what does your proof need to start with? (And what do you have to show about the thing you start with?)

D2. It isn't enough to just state the axioms—you need to explain how you are using them. (Giving names to all your points and lines can help with this.)

D3. You can't refer to objects you haven't yet defined.

D4. How do you know that these are different lines? (For example, in the following picture AB and AC are actually the same line.)



D5. How do you know these are different lines? Actually, they could be the same! On the sphere, if B and C are antipodal points on the equator L and point A is the north pole, then point A isn't on $L = BC$, but AB is still the same line as AC .

D6. This is one possible case, but what if your point is on L ?

D7. Can you make it clearer what the 2 cases are? What is it that distinguishes them?

D8. You only have to show that the point you started with has two lines through it.

D9. You've given an argument that some points have 2 lines through them, but you need to start with an arbitrary point and show that there are two different lines through it.

D10. I don't understand what you've done. You need to start with an arbitrary point, and using the axioms, one at a time, show how to find 2 different particular lines through that particular point.

D11. You need to start with a given point, and using the axioms, one at a time, show how to find a particular line that definitely doesn't go through that particular point.

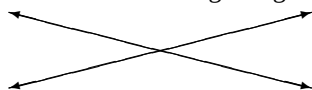
D12. You've given one example of a model in which the statement is true, but to be a

theorem, it must be true in all possible models. To do that, you need to give an argument using the axioms.

A.6 Vertical Angle Theorem Comments

E1. Is this a number? A shape? Something else?

E2. So is this 1 single angle?



According to your definition it seems to be.

E3. If you are saying that an angle is some kind of number, you need to say exactly how to find it from a figure.

E4. Your definition of congruence doesn't seem to match with your definition of an angle.

E5. So this is a number?

E6. Why?? How do you know this will always happen? If you are using a symmetry as part of your proof, you need to prove that the figure has the symmetry. It isn't enough to just say that it has the symmetry; you need to give a convincing argument that the figure will always have this symmetry.

E7. If you are saying that an angle is a geometric shape in the plane, you need to give a definition that is clear enough that I can tell exactly which points are part of the angle. (Is it made of rays, or line segments, or points between them, or something else...?)

E8. If you bisect an angle, you can't assume that you've also bisected its vertical angle. (To show this, you would need to use the vertical angle theorem, so your argument would be circular.)

E9. Is an angle a number, or a shape? If a shape, what does it mean to add them together? If a number, how exactly does the number relate to the shape? Often, people find it useful to distinguish between an angle and its measurement.

E10. This is the high school proof of the Vertical Angle Theorem that I asked people not to give, because it is difficult to fill in all the details of. If you want to give this proof, you need to give a careful explanation of how to relate angles as numbers with angles as shapes, and use that definition to prove the properties of angles that you are using.

A.7 SAS on the Sphere Comments

F1. You've given convincing counterexamples showing that Side-Angle-Side isn't true on the sphere, and said how you would change the statement of the theorem, but you still need to give a proof showing that your modified version always works on the sphere. Otherwise, there might still be other counterexamples.

F2. You've talked about how to change the theorem, but can you explicitly state your new, modified theorem?

F3. It isn't impossible on the sphere!

F4. You've given Euclid's proof essentially without change; but if Euclid's proof was correct without change on the sphere, then the original theorem would be true without change, too. So you need to figure out how to change the proof, as well as the theorem. Furthermore, your proof will have to use the part of the theorem that you changed in some crucial way.

F5. I don't understand this. Can you explain your counterexample(s) in greater detail?

F6. Your two triangles don't seem to have the same given interior angle.

F7. Your changes should modify the premise (the if part) of the theorem, not its conclusion.

F8. Your proof didn't use your changes to the statement of the theorem, but we know that Side-Angle-Side isn't true without these changes. So something must still be missing from your proof.

F9. Why does this force the two sides to coincide?

F10. What if there is more than one possible base that is the same (shortest) length?

A.8 Computer Lab II (Billiards Tables) Comments

G1. I don't understand this. How do you know this gives $4n$ paths?

G2. How do you construct the path after that?

G3. Why? How do you know that this always works? Why does it make the angle at which the ball hits the wall the same as the angle at which it leaves the wall?

G4. How do you construct the path of a ball after it's been hit? (That is, without worrying about hitting another ball.)

Appendix B

From *Flatland* by Edwin Abbott

This section is excerpted from [Abbott, 1885].

B.1 Part I: This World

“Be patient, for the world is broad and wide.”

Section 1: Of the nature of Flatland

I call our world Flatland, not because we call it so, but to make its nature clearer to you, my happy readers, who are privileged to live in Space.

Imagine a vast sheet of paper on which straight Lines, Triangles, Squares, Pentagons, Hexagons, and other figures, instead of remaining fixed in their places, move freely about, on or in the surface, but without the power of rising above or sinking below it, very much like shadows — only hard with luminous edges — and you will then have a pretty correct notion of my country and countrymen. Alas, a few years ago, I should have said “my universe”: but now my mind has been opened to higher views of things.

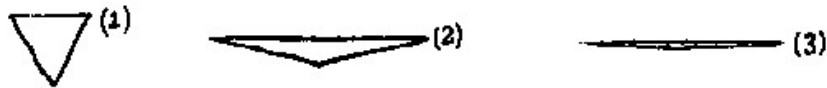
In such a country, you will perceive at once that it is impossible that there should be anything of what you call a “solid” kind; but I dare say you will suppose that we could at least distinguish by sight the Triangles, Squares, and other figures, moving about as I have described them. On the contrary, we could see nothing of the kind, not at least so as to distinguish one figure from another. Nothing was visible, nor could be visible, to us, except Straight Lines; and the necessity of this I will speedily demonstrate.

Place a penny on the middle of one of your tables in Space; and leaning over it, look down upon it. It will appear a circle.

But now, drawing back to the edge of the table, gradually lower your eye (thus bringing yourself more and more into the condition of the inhabitants of Flatland), and you will find the penny becoming more and more oval to your view, and at last when you have placed your eye exactly on the edge of the table (so that you are, as it were, actually a Flatlander) the penny will then have ceased to appear oval at all, and will have become, so far as you can see, a straight line.

The same thing would happen if you were to treat in the same way a Triangle, or a Square, or any other figure cut out from pasteboard. As soon as you look at it with your eye on the edge of the table, you will find that it ceases to appear to you as a figure, and that it becomes in appearance a straight line. Take for example an equilateral Triangle

— who represents with us a Tradesman of the respectable class. Figure 1 represents the Tradesman as you would see him while you were bending over him from above; figures 2 and 3 represent the Tradesman, as you would see him if your eye were close to the level, or all but on the level of the table; and if your eye were quite on the level of the table (and that is how we see him in Flatland) you would see nothing but a straight line.



When I was in Spaceland I heard that your sailors have very similar experiences while they traverse your seas and discern some distant island or coast lying on the horizon. The far-off land may have bays, forelands, angles in and out to any number and extent; yet at a distance you see none of these (unless indeed your sun shines bright upon them revealing the projections and retirements by means of light and shade), nothing but a grey unbroken line upon the water.

Well, that is just what we see when one of our triangular or other acquaintances comes towards us in Flatland. As there is neither sun with us, nor any light of such a kind as to make shadows, we have none of the helps to the sight that you have in Spaceland. If our friend comes closer to us we see his line becomes larger; if he leaves us it becomes smaller; but still he looks like a straight line; be he a Triangle, Square, Pentagon, Hexagon, Circle, what you will — a straight Line he looks and nothing else.

You may perhaps ask how under these disadvantages circumstances we are able to distinguish our friends from one another: but the answer to this very natural question will be more fitly and easily given when I come to describe the inhabitants of Flatland. For the present let me defer this subject, and say a word or two about the climate and houses in our country.

Section 2: Of the climate and houses in Flatland

As with you, so also with us, there are four points of the compass North, South, East, and West.

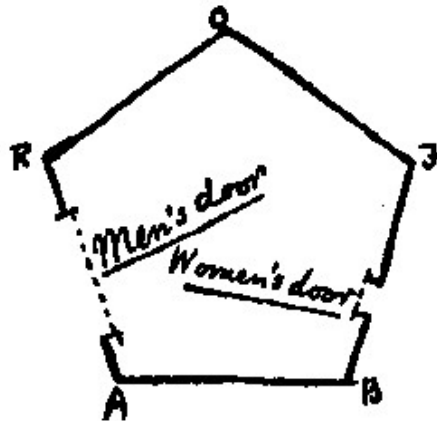
There being no sun nor other heavenly bodies, it is impossible for us to determine the North in the usual way; but we have a method of our own. By a Law of Nature with us, there is a constant attraction to the South; and, although in temperate climates this is very slight — so that even a Woman in reasonable health can journey several furlongs northward without much difficulty — yet the hampering effort of the southward attraction is quite sufficient to serve as a compass in most parts of our earth. Moreover, the rain (which falls at stated intervals) coming always from the North, is an additional assistance; and in the towns we have the guidance of the houses, which of course have their side-walls running for the most part North and South, so that the roofs may keep off the rain from the North. In the country, where there are no houses, the trunks of the trees serve as some sort of guide. Altogether, we have not so much difficulty as might be expected in determining our bearings.

Yet in our more temperate regions, in which the southward attraction is hardly felt, walking sometimes in a perfectly desolate plain where there have been no houses nor trees to guide me, I have been occasionally compelled to remain stationary for hours together, waiting till the rain came before continuing my journey. On the weak and aged, and especially on delicate Females, the force of attraction tells much more heavily than on the robust of the Male Sex, so that it is a point of breeding, if you meet a Lady on the street,

always to give her the North side of the way — by no means an easy thing to do always at short notice when you are in rude health and in a climate where it is difficult to tell your North from your South.

Windows there are none in our houses: for the light comes to us alike in our homes and out of them, by day and by night, equally at all times and in all places, whence we know not. It was in old days, with our learned men, an interesting and oft-investigate question, “What is the origin of light?” and the solution of it has been repeatedly attempted, with no other result than to crowd our lunatic asylums with the would-be solvers. Hence, after fruitless attempts to suppress such investigations indirectly by making them liable to a heavy tax, the Legislature, in comparatively recent times, absolutely prohibited them. I — alas, I alone in Flatland — know now only too well the true solution of this mysterious problem; but my knowledge cannot be made intelligible to a single one of my countrymen; and I am mocked at — I, the sole possessor of the truths of Space and of the theory of the introduction of Light from the world of three Dimensions — as if I were the maddest of the mad! But a truce to these painful digressions: let me return to our homes.

The most common form for the construction of a house is five-sided or pentagonal, as in the annexed figure. The two Northern sides RO, OF, constitute the roof, and for the most part have no doors; on the East is a small door for the Women; on the West a much larger one for the Men; the South side or floor is usually doorless.



Square and triangular houses are not allowed, and for this reason. The angles of a Square (and still more those of an equilateral Triangle,) being much more pointed than those of a Pentagon, and the lines of inanimate objects (such as houses) being dimmer than the lines of Men and Women, it follows that there is no little danger lest the points of a square or triangular house residence might do serious injury to an inconsiderate or perhaps absentminded traveller suddenly running against them: and therefore, as early as the eleventh century of our era, triangular houses were universally forbidden by Law, the only exceptions being fortifications, powder-magazines, barracks, and other state buildings, which is not desirable that the general public should approach without circumspection.

At this period, square houses were still everywhere permitted, though discouraged by a special tax. But, about three centuries afterwards, the Law decided that in all towns containing a population above ten thousand, the angle of a Pentagon was the smallest house-angle that could be allowed consistently with the public safety. The good sense of the community has seconded the efforts of the Legislature; and now, even in the country, the pentagonal construction has superseded every other. It is only now and then in some very remote and backward agricultural district that an antiquarian may still discover a square house.

Section 3: Concerning the inhabitants of Flatland

The greatest length or breadth of a full grown inhabitant of Flatland may be estimated at about eleven of your inches. Twelve inches may be regarded as a maximum.

Our Women are Straight Lines.

Our Soldiers and Lowest Class of Workmen are Triangles with two equal sides, each about eleven inches long, and a base or third side so short (often not exceeding half an inch) that they form at their vertices a very sharp and formidable angle. Indeed when their bases are of the most degraded type (not more than the eighth part of an inch in size), they can hardly be distinguished from Straight lines or Women; so extremely pointed are their vertices. With us, as with you, these Triangles are distinguished from others by being called Isosceles; and by this name I shall refer to them in the following pages.

Our Middle Class consists of Equilateral or Equal-Sided Triangles.

Our Professional Men and Gentlemen are Squares (to which class I myself belong) and Five-Sided Figures or Pentagons.

Next above these come the Nobility, of whom there are several degrees, beginning at Six-Sided Figures, or Hexagons, and from thence rising in the number of their sides till they receive the honourable title of Polygonal, or many-Sided. Finally when the number of the sides becomes so numerous, and the sides themselves so small, that the figure cannot be distinguished from a circle, he is included in the Circular or Priestly order; and this is the highest class of all.

It is a Law of Nature with us that a male child shall have one more side than his father, so that each generation shall rise (as a rule) one step in the scale of development and nobility. Thus the son of a Square is a Pentagon; the son of a Pentagon, a Hexagon; and so on.

But this rule applies not always to the Tradesman, and still less often to the Soldiers, and to the Workmen; who indeed can hardly be said to deserve the name of human Figures, since they have not all their sides equal. With them therefore the Law of Nature does not hold; and the son of an Isosceles (*i.e.*, a Triangle with two sides equal) remains Isosceles still. Nevertheless, all hope is not such out, even from the Isosceles, that his posterity may ultimately rise above his degraded condition. For, after a long series of military successes, or diligent and skillful labours, it is generally found that the more intelligent among the Artisan and Soldier classes manifest a slight increase of their third side or base, and a shrinkage of the two other sides. Intermarriages (arranged by the Priests) between the sons and daughters of these more intellectual members of the lower classes generally result in an offspring approximating still more to the type of the Equal-Sided Triangle.

Rarely — in proportion to the vast numbers of Isosceles births — is a genuine and certifiable Equal-Sided Triangle produced from Isosceles parents.¹ Such a birth requires, as its antecedents, not only a series of carefully arranged intermarriages, but also a long-continued exercise of frugality and self-control on the part of the would-be ancestors of the coming Equilateral, and a patient, systematic, and continuous development of the Isosceles intellect through many generations.

The birth of a True Equilateral Triangle from Isosceles parents is the subject of rejoicing in our country for many furlongs round. After a strict examination conducted by the Sanitary and Social Board, the infant, if certified as Regular, is with solemn ceremonial

¹“What need of a certificate?” a Spaceland critic may ask: “Is not the procreation of a Square Son a certificate from Nature herself, proving the Equal-sidedness of the Father?” I reply that no Lady of any position will marry an uncertified Triangle. Square offspring has sometimes resulted from a slightly Irregular Triangle; but in almost every such case the Irregularity of the first generation is visited on the third; which either fails to attain the Pentagonal rank, or relapses to the Triangular.

admitted into the class of Equilaterals. He is then immediately taken from his proud yet sorrowing parents and adopted by some childless Equilateral, who is bound by oath never to permit the child henceforth to enter his former home or so much as to look upon his relations again, for fear lest the freshly developed organism may, by force of unconscious imitation, fall back again into his hereditary level.

The occasional emergence of an Equilateral from the ranks of his serf-born ancestors is welcomed, not only by the poor serfs themselves, as a gleam of light and hope shed upon the monotonous squalor of their existence, but also by the Aristocracy at large; for all the higher classes are well aware that these rare phenomena, while they do little or nothing to vulgarize their own privileges, serve as almost useful barrier against revolution from below.

Had the acute-angled rabble been all, without exception, absolutely destitute of hope and of ambition, they might have found leaders in some of their many seditious outbreaks, so able as to render their superior numbers and strength too much even for the wisdom of the Circles. But a wise ordinance of Nature has decreed that, in proportion as the working-classes increase in intelligence, knowledge, and all virtue, in that same proportion their acute angle (which makes them physically terrible) shall increase also and approximate to their comparatively harmless angle of the Equilateral Triangle. Thus, in the most brutal and formidable off the soldier class — creatures almost on a level with women in their lack of intelligence — it is found that, as they wax in the mental ability necessary to employ their tremendous penetrating power to advantage, so do they wane in the power of penetration itself.

How admirable is the Law of Compensation! And how perfect a proof of the natural fitness and, I may almost say, the divine origin of the aristocratic constitution of the States of Flatland! By a judicious use of this Law of Nature, the Polygons and Circles are almost always able to stifle sedition in its very cradle, taking advantage of the irrepressible and boundless hopefulness of the human mind. Art also comes to the aid of Law and Order. It is generally found possible — by a little artificial compression or expansion on the part of the State physicians — to make some of the more intelligent leaders of a rebellion perfectly Regular, and to admit them at once into the privileged classes; a much larger number, who are still below the standard, allured by the prospect of being ultimately ennobled, are induced to enter the State Hospitals, where they are kept in honourable confinement for life; one or two alone of the most obstinate, foolish, and hopelessly irregular are led to execution.

Then the wretched rabble of the Isosceles, planless and leaderless, are ether transfixed without resistance by the small body of their brethren whom the Chief Circle keeps in pay for emergencies of this kind; or else more often, by means of jealousies and suspicious skillfully fomented among them by the Circular party, they are stirred to mutual warfare, and perish by one another's angles. No less than one hundred and twenty rebellions are recorded in our annals, besides minor outbreaks numbered at two hundred and thirty-five; and they have all ended thus.

Section 4: Concerning the women

If our highly pointed Triangles of the Soldier class are formidable, it may be readily inferred that far more formidable are our Women. For, if a Soldier is a wedge, a Woman is a needle; being, so to speak, all point, at least at the two extremities. Add to this the power of making herself practically invisible at will, and you will perceive that a Female, in Flatland, is a creature by no means to be trifled with.

But here, perhaps, some of my younger Readers may ask how a woman in Flatland

can make herself invisible. This ought, I think, to be apparent without any explanation. However, a few words will make it clear to the most unreflecting.

Place a needle on the table. Then, with your eye on the level of the table, look at it side-ways, and you see the whole length of it; but look at it end-ways, and you see nothing but a point, it has become practically invisible. Just so is it with one of our Women. When her side is turned towards us, we see her as a straight line; when the end containing her eye or mouth — for with us these two organs are identical — is the part that meets our eye, then we see nothing but a highly lustrous point; but when the back is presented to our view, then — being only sub-lustrous, and, indeed, almost as dim as an inanimate object — her hinder extremity serves her as a kind of Invisible Cap.

The dangers to which we are exposed from our Women must now be manifest to the meanest capacity of Spaceland. If even the angle of a respectable Triangle in the middle class is not without its dangers; if to run against a Working Man involves a gash; if collision with an Officer of the military class necessitates a serious wound; if a mere touch from the vertex of a Private Soldier brings with it danger of death; — what can it be to run against a woman, except absolute and immediate destruction? And when a Woman is invisible, or visible only as a dim sub-lustrous point, how difficult must it be, even for the most cautious, always to avoid collision!

Many are the enactments made at different times in the different States of Flatland, in order to minimize this peril; and in the Southern and less temperate climates, where the force of gravitation is greater, and human beings more liable to casual and involuntary motions, the Laws concerning Women are naturally much more stringent. But a general view of the Code may be obtained from the following summary: —

1. Every house shall have one entrance on the Eastern side, for the use of Females only; by which all females shall enter “in a becoming and respectful manner”² and not by the Men’s or Western door.
2. No Female shall walk in any public place without continually keeping up her Peace-cry, under penalty of death.
3. Any Female, duly certified to be suffering from St. Vitus’s Dance, fits, chronic cold accompanied by violent sneezing, or any disease necessitating involuntary motions, shall be instantly destroyed.

In some of the States there is an additional Law forbidding Females, under penalty of death, from walking or standing in any public place without moving their backs constantly from right to left so as to indicate their presence to those behind them; other oblige a Woman, when travelling, to be followed by one of her sons, or servants, or by her husband; others confine Women altogether in their houses except during the religious festivals. But it has been found by the wisest of our Circles or Statesmen that the multiplication of restrictions on Females tends not only to the debilitation and diminution of the race, but also to the increase of domestic murders to such an extent that a State loses more than it gains by a too prohibitive Code.

For whenever the temper of the Women is thus exasperated by confinement at home or hampering regulations abroad, they are apt to vent their spleen upon their husbands and children; and in the less temperate climates the whole male population of a village has been sometimes destroyed in one or two hours of a simultaneous female outbreak. Hence the

²When I was in Spaceland I understood that some of your Priestly circles have in the same way a separate entrance for Villagers, Farmers and Teachers of Board Schools (Spectator, Sept. 1884, p. 1255) that they may “approach in a becoming and respectful manner.”

Three Laws, mentioned above, suffice for the better regulated States, and may be accepted as a rough exemplification of our Female Code.

After all, our principal safeguard is found, not in Legislature, but in the interests of the Women themselves. For, although they can inflict instantaneous death by a retrograde movement, yet unless they can at once disengage their stinging extremity from the struggling body of their victim, their own frail bodies are liable to be shattered.

The power of Fashion is also on our side. I pointed out that in some less civilized States no female is suffered to stand in any public place without swaying her back from right to left. This practice has been universal among ladies of any pretensions to breeding in all well-governed States, as far back as the memory of Figures can reach. It is considered a disgrace to any state that legislation should have to enforce what ought to be, and is in every respectable female, a natural instinct. The rhythmical and, if I may so say, well-modulated undulation of the back in our ladies of Circular rank is envied and imitated by the wife of a common Equilateral, who can achieve nothing beyond a mere monotonous swing, like the ticking of a pendulum; and the regular tick of the Equilateral is no less admired and copied by the wife of the progressive and aspiring Isosceles, in the females of whose family no “back-motion” of any kind has become as yet a necessity of life. Hence, in every family of position and consideration, “back motion” is as prevalent as time itself; and the husbands and sons in these households enjoy immunity at least from invisible attacks.

Not that it must be for a moment supposed that our Women are destitute of affection. But unfortunately the passion of the moment predominates, in the Frail Sex, over every other consideration. This is, of course, a necessity arising from their unfortunate conformation. For as they have no pretensions to an angle, being inferior in this respect to the very lowest of the Isosceles, they are consequently wholly devoid of brainpower, and have neither reflection, judgment nor forethought, and hardly any memory. Hence, in their fits of fury, they remember no claims and recognize no distinctions. I have actually known a case where a Woman has exterminated her whole household, and half an hour afterwards, when her rage was over and the fragments swept away, has asked what has become of her husband and children.

Obviously then a Woman is not to be irritated as long as she is in a position where she can turn round. When you have them in their apartments — which are constructed with a view to denying them that power — you can say and do what you like; for they are then wholly impotent for mischief, and will not remember a few minutes hence the incident for which they may be at this moment threatening you with death, nor the promises which you may have found it necessary to make in order to pacify their fury.

On the whole we got on pretty smoothly in our domestic relations, except in the lower strata of the Military Classes. There the want of tact and discretion on the part of the husbands produces at times indescribable disasters. Relying too much on the offensive weapons of their acute angles instead of the defensive organs of good sense and seasonable simulations, these reckless creatures too often neglect the prescribed construction of the women’s apartments, or irritate their wives by ill-advised expressions out of doors, which they refuse immediately to retract. Moreover a blunt and stolid regard for literal truth indisposes them to make those lavish promises by which the more judicious Circle can in a moment pacify his consort. The result is massacre; not, however, without its advantages, as it eliminates the more brutal and troublesome of the Isosceles; and by many of our Circles the destructiveness of the Thinner Sex is regarded as one among many providential arrangements for suppressing redundant population, and nipping Revolution in the bud.

Yet even in our best regulated and most approximately Circular families I cannot say that the ideal of family life is so high as with you in Spaceland. There is peace, in so far as

the absence of slaughter may be called by that name, but there is necessarily little harmony of tastes or pursuits; and the cautious wisdom of the Circles has ensured safety at the cost of domestic comfort. In every Circular or Polygonal household it has been a habit from time immemorial — and now has become a kind of instinct among the women of our higher classes — that the mothers and daughters should constantly keep their eyes and mouths towards their husband and his male friends; and for a lady in a family of distinction to turn her back upon her husband would be regarded as a kind of portent, involving loss of status. But, as I shall soon shew, this custom, though it has the advantage of safety, is not without disadvantages.

In the house of the Working Man or respectable Tradesman — where the wife is allowed to turn her back upon her husband, while pursuing her household avocations — there are at least intervals of quiet, when the wife is neither seen nor heard, except for the humming sound of the continuous Peace-cry; but in the homes of the upper classes there is too often no peace. There the voluble mouth and bright penetrating eye are ever directed toward the Master of the household; and light itself is not more persistent than the stream of Feminine discourse. The tact and skill which suffice to avert a Woman's sting are unequal to the task of stopping a Woman's mouth; and as the wife has absolutely nothing to say, and absolutely no constraint of wit, sense, or conscience to prevent her from saying it, not a few cynics have been found to aver that they prefer the danger of the death-dealing but inaudible sting to the safe sonorousness of a Woman's other end.

To my readers in Spaceland the condition of our Women may seem truly deplorable, and so indeed it is. A Male of the lowest type of the Isosceles may look forward to some improvement of his angle, and to the ultimate elevation of the whole of his degraded caste; but no Woman can entertain such hopes for her sex. "Once a Woman, always a Woman" is a Decree of Nature; and the very Laws of Evolution seem suspended in her disfavour. Yet at least we can admire the wise Prearrangement which has ordained that, as they have no hopes, so they shall have no memory to recall, and no forethought to anticipate, the miseries and humiliations which are at once a necessity of their existence and the basis of the constitution of Flatland.

Section 5: Of our methods of recognizing one another

You, who are blessed with shade as well as light, you, who are gifted with two eyes, endowed with a knowledge of perspective, and charmed with the enjoyment of various colours, you, who can actually see an angle, and contemplate the complete circumference of a Circle in the happy region of the Three Dimensions — how shall I make it clear to you the extreme difficulty which we in Flatland experience in recognizing one another's configuration?

Recall what I told you above. All beings in Flatland, animate and inanimate, no matter what their form, present to our view the same, or nearly the same, appearance, viz. that of a straight Line. How then can one be distinguished from another, where all appear the same?

The answer is threefold. The first means of recognition is the sense of hearing; which with us is far more highly developed than with you, and which enables us not only to distinguish by the voice of our personal friends, but even to discriminate between different classes, at least so far as concerns the three lowest orders, the Equilateral, the Square, and the Pentagon — for the Isosceles I take no account. But as we ascend the social scale, the process of discriminating and being discriminated by hearing increases in difficulty, partly because voices are assimilated, partly because the faculty of voice-discrimination is a plebeian virtue not much developed among the Aristocracy. And wherever there is any danger of imposture we cannot trust to this method. Amongst our lowest orders, the vocal

organs are developed to a degree more than correspondent with those of hearing, so that an Isosceles can easily feign the voice of a Polygon, and, with some training, that of a Circle himself. A second method is therefore more commonly resorted to.

Feeling is, among our Women and lower classes — about our upper classes I shall speak presently — the principal test of recognition, at all events between strangers, and when the question is, not as to the individual, but as to the class. What therefore “introduction” is among the higher classes in Spaceland, that the process of “feeling” is with us. “Permit me to ask you to feel and be felt by my friend Mr. So-and-so” — is still, among the more old-fashioned of our country gentlemen in districts remote from towns, the customary formula for a Flatland introduction. But in the towns, and among men of business, the words “be felt by” are omitted and the sentence is abbreviated to, “Let me ask you to feel Mr. So-and-so”; although it is assumed, of course, that the “feeling” is to be reciprocal. Among our still more modern and dashing young gentlemen — who are extremely averse to superfluous effort and supremely indifferent to the purity of their native language — the formula is still further curtailed by the use of “to feel” in a technical sense, meaning, “to recommend-for-the-purposes-of-feeling-and-being-felt”; and at this moment the “slang” of polite or fast society in the upper classes sanctions such a barbarism as “Mr. Smith, permit me to feel Mr. Jones.”

Let not my Reader however suppose that “feeling” is with us the tedious process that it would be with you, or that we find it necessary to feel right round all the sides of every individual before we determine the class to which he belongs. Long practice and training, begun in the schooles and continued in the experience of daily life, enable us to discriminate at once by the sense of touch, between the angles of an equal-sided Triangle, Square, and Pentagon; and I need not say that the brainless vertex of an acute-angled Isosceles is obvious to the dumbest touch. It is therefore not necessary, as a rule, to do more than feel a single angle of an individual; and this, once ascertained, tells us the class of the person whom we are addressing, unless indeed he belongs to the higher sections of the nobility. There the difficulty is much greater. Even a Master of Arts in our University of Wentbridge has been known to confuse a ten-sided with a twelve-sided Polygon; and there is hardly a Doctor of Science in or out of that famous University who could pretend to decide promptly and unhesitatingly between a twenty-sided and a twenty-four sided member of the Aristocracy.

Those of my readers who recall the extracts I gave above from the Legislative code concerning Women, will readily perceive that the process of introduction by contact requires some care and discretion. Otherwise the angles might inflict on the unwary Feeling irreparable injury. It is essential for the safety of the Feeler that the Felt should stand perfectly still. A start, a fidgety shifting of the position, yes, even a violent sneeze, has been known before now to prove fatal to the incautious, and to nip in the bud many a promising friendship. Especially is this true among the lower classes of the Triangles. With them, the eye is situated so far from their vertex that they can scarcely take cognizance of what goes on at that extremity of their frame. They are, moreover, of a rough coarse nature, not sensitive to the delicate touch of the highly organized Polygon. What wonder then if an involuntary toss of the head has ere now deprived the State of a valuable life!

I have heard that my excellent Grandfather — one of the least irregular of his unhappy Isosceles class, who indeed obtained, shortly before his decease, four out of seven votes from the Sanitary and Social Board for passing him into the class of the Equal-sided — often deplored, with a tear in his venerable eye, a miscarriage of this kind, which had occurred to his great-great-great-Grandfather, a respectable Working Man with an angle or brain of 59 degrees 30 minutes. According to his account, my unfortunately Ancestor, being afflicted with rheumatism, and in the act of being felt by a Polygon, by one sudden start

accidentally transfixed the Great Man through the diagonal and thereby, partly in consequence of his long imprisonment and degradation, and partly because of the moral shock which pervaded the whole of my Ancestor's relations, threw back our family a degree and a half in their ascent towards better things. The result was that in the next generation the family brain was registered at only 58 degrees, and not till the lapse of five generations was the lost ground recovered, the full 60 degrees attained, and the Ascent from the Isosceles finally achieved. And all this series of calamities from one little accident in the process of Feeling.

As this point I think I hear some of my better educated readers exclaim, "How could you in Flatland know anything about angles and degrees, or minutes? We see an angle, because we, in the region of Space, can see two straight lines inclined to one another; but you, who can see nothing but on straight line at a time, or at all events only a number of bits of straight lines all in one straight line, — how can you ever discern an angle, and much less register angles of different sizes?"

I answer that though we cannot see angles, we can infer them, and this with great precision. Our sense of touch, stimulated by necessity, and developed by long training, enables us to distinguish angles far more accurately than your sense of sight, when unaided by a rule or measure of angles. nor must I omit to explain that we have great natural helps. It is with us a Law of Nature that the brain of the Isosceles class shall begin at half a degree, or thirty minutes, and shall increase (if it increases at all) by half a degree in every generation until the goal of 60 degrees is reached, when the condition of serfdom is quitted, and the freeman enters the class of Regulars.

Consequently, Nature herself supplies us with an ascending scale or Alphabet of angles for half a degree up to 60 degrees, Specimen of which are placed in every Elementary School throughout the land. Owing to occasional retrogressions, to still more frequent moral and intellectual stagnation, and to the extraordinary fecundity of the Criminal and Vagabond classes, there is always a vast superfluity of individuals of the half degree and single degree class, and a fair abundance of Specimens up to 10 degrees. These are absolutely destitute of civil rights; and a great number of them, not having even intelligence enough for the purposes of warfare, are devoted by the States to the service of education. Fettered immovably so as to remove all possibility of danger, they are placed in the classrooms of our Infant Schools, and there they are utilized by the Board of Education for the purpose of imparting to the offspring of the Middle Classes the tact and intelligence which these wretched creatures themselves are utterly devoid.

In some States the Specimens are occasionally fed and suffered to exist for several years; but in the more temperate and better regulated regions, it is found in the long run more advantageous for the educational interests of the young, to dispense with food, and to renew the Specimens every month — which is about the average duration of the foodless existence of the Criminal class. In the cheaper schools, what is gained by the longer existence of the Specimen is lost, partly in the expenditure for food, and partly in the diminished accuracy of the angles, which are impaired after a few weeks of constant "feeling." Nor must we forget to add, in enumerating the advantages of the more expensive system, that it tends, though slightly yet perceptibly, to the diminution of the redundant Isosceles population — an object which every statesman in Flatland constantly keeps in view. On the whole therefore — although I am not ignorant that, in many popularly elected School Boards, there is a reaction in favour of "the cheap system" as it is called — I am myself disposed to think that this is one of the many cases in which expense is the truest economy.

But I must not allow questions of School Board politics to divert me from my subject. Enough has been said, I trust, to shew that Recognition by Feeling is not so tedious or indecisive a process as might have been supposed; and it is obviously more trustwor-

thy than Recognition by hearing. Still there remains, as has been pointed out above, the objection that this method is not without danger. For this reason many in the Middle and Lower classes, and all without exception in the Polygonal and Circular orders, prefer a third method, the description of which shall be reserved for the next section.

Section 6: Of recognition by sight

I am about to appear very inconsistent. In the previous sections I have said that all figures in Flatland present the appearance of a straight line; and it was added or implied, that it is consequently impossible to distinguish by the visual organ between individuals of different classes: yet now I am about to explain to my Spaceland critics how we are able to recognize one another by the sense of sight.

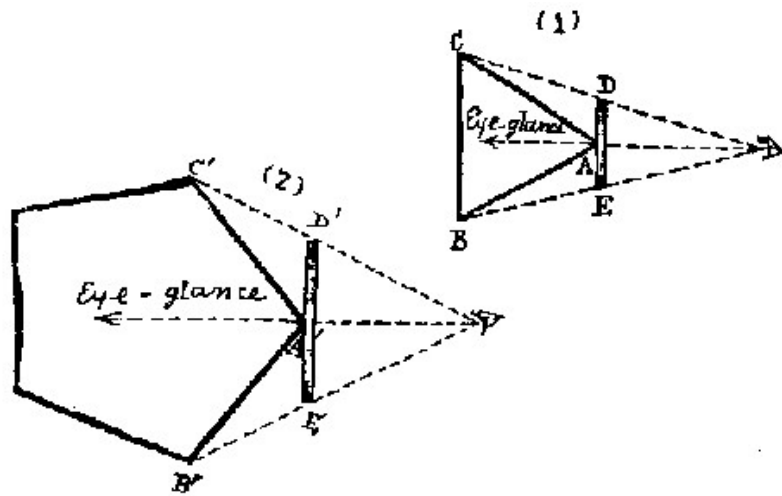
If however the Reader will take the trouble to refer to the passage in which Recognition by Feeling is stated to be universal, he will find this qualification – "among the lower classes." It is only among the higher classes and in our more temperate climates that Sight Recognition is practised.

That this power exists in any regions and for any classes is the result of Fog; which prevails during the greater part of the year in all parts save the torrid zones. That which is with you in Spaceland an unmixed evil, blotting out the landscape, depressing the spirits, and enfeebling the health, is by us recognized as a blessing scarcely inferior to air itself, and as the Nurse of arts and Parent of sciences. But let me explain my meaning, without further eulogies on this beneficent Element.

If Fog were non-existent, all lines would appear equally and indistinguishably clear; and this is actually the case in those unhappy countries in which the atmosphere is perfectly dry and transparent. But wherever there is a rich supply of Fog, objects that are at a distance, say of three feet, are appreciably dimmer than those at the distance of two feet eleven inches; and the result is that by careful and constant experimental observation of comparative dimness and constant experimental observation of comparative dimness and clearness, we are enabled to infer with great exactness the configuration of the object observed.

An instance will do more than a volume of generalities to make my meaning clear.

Suppose I see two individuals approaching whose rank I wish to ascertain. They are, we will suppose, a Merchant and a Physician, or in other words, an Equilateral Triangle and a Pentagon; how am I to distinguish them?



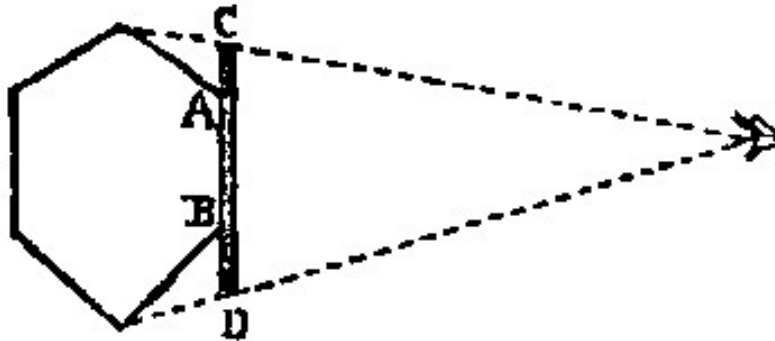
It will be obvious, to every child in Spaceland who has touched the threshold of Geometrical Studies, that, if I can bring my eye so that its glance may bisect an angle (A) of the approaching stranger, my view will lie as it were evenly between the two sides that are next to me (viz. CA and AB), so that I shall contemplate the two impartially, and both will appear of the same size.

Now in the case of (1) the Merchant, what shall I see? I shall see a straight line DAE, in which the middle point (A) will be very bright because it is nearest to me; but on either side the line will shade away rapidly to dimness, because the sides AC and AB recede rapidly into the fog and what appear to me as the Merchant's extremities, viz. D and E, will be very dim indeed.

On the other hand in the case of (2) the Physician, though I shall here also see a line (D'A'E') with a bright centre (A'), yet it will shade away less rapidly to dimness, because the sides (A'C', A'B') recede less rapidly into the fog: and what appear to me the Physician's extremities, viz. D' and E', will not be not so dim as the extremities of the Merchant.

The Reader will probably understand from these two instances how -- after a very long training supplemented by constant experience -- it is possible for the well-educated classes among us to discriminate with fair accuracy between the middle and lowest orders, by the sense of sight. If my Spaceland Patrons have grasped this general conception, so far as to conceive the possibility of it and not to reject my account as altogether incredible -- I shall have attained all I can reasonably expect. Were I to attempt further details I should only perplex. Yet for the sake of the young and inexperienced, who may perchance infer -- from the two simple instances I have given above, of the manner in which I should recognize my Father and my Sons -- that Recognition by sight is an easy affair, it may be needful to point out that in actual life most of the problems of Sight Recognition are far more subtle and complex.

If for example, when my Father, the Triangle, approaches me, he happens to present his side to me instead of his angle, then, until I have asked him to rotate, or until I have edged my eye around him, I am for the moment doubtful whether he may not be a Straight Line, or, in other words, a Woman. Again, when I am in the company of one of my two hexagonal Grandsons, contemplating one of his sides (AB) full front, it will be evident from the accompanying diagram that I shall see one whole line (AB) in comparative brightness (shading off hardly at all at the ends) and two smaller lines (CA and BD) dim throughout and shading away into greater dimness towards the extremities C and D.



But I must not give way to the tempting of enlarging on these topics. The meanest mathematician in Spaceland will readily believe me when I assert that the problems of life, which present themselves to the well-educated – when they are themselves in motion, rotating, advancing or retreating, and at the same time attempting to discriminate by the sense of sight between a number of Polygons of high rank moving in different directions, as for example in a ball-room or conversazione – must be of a nature to task the angularity of the most intellectual, and amply justify the rich endowments of the Learned Professors of Geometry, both Static and Kinetic, in the illustrious University of Wentbridge, where the Science and Art of Sight Recognition are regularly taught to large classes of the elite of the States.

It is only a few of the scions of our noblest and wealthies houses, who are able to give the time and money necessary for the thorough prosecution of this noble and valuable Art. Even to me, a Mathematician of no mean standing, and the Grandfather of two most hopeful and perfectly regular Hexagons, to find myself in the midst of a crowd of rotating Polygons of the higher classes, is occasionally very perplexing. And of course to a common Tradesman, or Serf, such a sight is almost as unintelligible as it would be to you, my Reader, were you suddenly transported to my country.

In such a crowd you could see on all sides of you nothing but a Line, apparently straight, but of which the parts would vary irregularly and perpetually in brightness or dimness. Even if you had completed your third year in the Pentagonal and Hexagonal classes in the University, and were perfect in the theory of the subject, you would still find there was need of many years of experience, before you could move in a fashionable crowd without jostling against your betters, whom it is against etiquette to ask to "feel," and who, by their superior culture and breeding, know all about your movements, while you know very little or nothing about theirs. in a word, to comport oneself with perfect propriety in Polygonal society, one ought to be a Polygon oneself. Such at least is the painful teaching of my experience.

It is astonishing how much the Art — or I may almost call it instinct — of Sight Recognition is developed by the habitual practice of it and by the avoidance of the custom of "Feeling." Just as, with you, the deaf and dumb, if once allowed to gesticulate and to use the hand-alphabet, will never acquire the more difficult but far more valuable art of lip-speech and lip-reading, so it is with us as regards "Seeing" and "Feeling." None who in early life resort to "Feeling" will ever learn "Seeing" in perfection.

For this reason, among our Higher Classes, "Feeling" is discouraged or absolutely forbidden. From the cradle their children, instead of going to the Public Elementary schools (where the art of Feeling is taught,) are sent to higher Seminaries of an exclusive character; and at our illustrious University, to "feel" is regarded as a most serious fault, involving Rustication for the first offence, and Expulsion for the second.

But among the lower classes the art of Sight Recognition is regarded as an unattainable luxury. A common Tradesman cannot afford to let his son spend a third of his life in abstract studies. The children of the poor are therefore allowed to "feel" from their earliest years, and they gain thereby a precocity and an early vivacity which contrast at first most favourably with the inert, undeveloped, and listless behaviour of the half-instructed youths of the Polygonal class; but when the latter have at last completed their University course, and are prepared to put their theory into practice, the change that comes over them may almost be described as a new birth, and in every art, science, and social pursuit they rapidly overtake and distance their Triangular competitors.

Only a few of the Polygonal Class fail to pass the Final Test or Leaving Examination at the University. The condition of the unsuccessful minority is truly pitiable. Rejected from the higher class, they are also despised by the lower. They have neither the matured and systematically trained powers of the Polygonal Bachelors and Masters of Arts, nor yet the native precocity and mercurial versatility of the youthful Tradesman. The professions, the public services, are closed against them, and though in most States they are not actually debarred from marriage, yet they have the greatest difficulty in forming suitable alliances, as experience shews that the offspring of such unfortunate and ill-endowed parents is generally itself unfortunate, if not positively Irregular.

It is from these specimens of the refuse of our Nobility that the great Tumults and Seditions of past ages have generally derived their leaders; and so great is the mischief thence arising that an increasing minority of our more progressive Statesmen are of opinion that true mercy would dictate their entire suppression, by enacting that all who fail to pass the Final Examination of the University should be either imprisoned for life, or extinguished by a painless death.

But I find myself digressing into the subject of Irregularities, a matter of such vital interest that it demands a separate section.

Appendix C

From Book I of Euclid's *Elements*

Euclid wrote the thirteen books of the *Elements* around 300 B.C. The first four books are about elementary planar geometry; books five through ten are largely about the theory of ratio and proportion and about what would now be considered number theory; and books eleven through thirteen are about three dimensional geometry.

The excerpt reprinted here is taken from the edition translated and edited by Thomas Heath [Euclid, 1956].

C.1 Definitions

1. A **point** is that which has no part.
2. A **line** is breadthless length.
3. The extremities of a line are points.
4. A **straight** line is a line which lies evenly with the points on itself.
5. A **surface** is that which has length and breadth only.
6. The extremities of a surface are lines.
7. A **plane** surface is a surface which lies evenly with the straight lines on itself.
8. A **plane angle** is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.
9. And when the lines containing the angle are straight, the angle is called **rectilineal**.
10. When a straight line set up on a straight line makes the adjacent angles equal to one another, each of the equal angles is **right**, and the straight line standing on the other is called a **perpendicular** to that on which it stands.
11. An **obtuse** angle is an angle greater than a right angle.
12. An **acute** angle is an angle less than a right angle.
13. A **boundary** is that which is an extremity of anything.
14. A **figure** is that which is contained by any boundary or boundaries.

15. A **circle** is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure are equal to one another;
16. And the point is called the **centre** of the circle.
17. A **diameter** of the circle is any straight line drawn through the centre and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.
18. A **semicircle** is the figure contained by the diameter and the circumference cut off by it. And the centre of the semicircle is the same as that of the circle.
19. **Rectilineal figures** are those which are contained by straight lines, **trilateral** figures being those contained by three, **quadrilateral** those contained by four, and **multilateral** those contained by more than four straight lines.
20. Of trilateral figures, an **equilateral** triangle is that which has its three sides equal, an **isosceles** triangle that which has two of its sides alone equal, and a **scalene** triangle that which has its three sides unequal.
21. Further, of trilateral figures, a **right-angled** triangle is that which has a right angle, an **obtuse-angled** triangle that which has an obtuse angle, and an **acute-angled** triangle that which has its three angles acute.
22. Of quadrilateral figures, a **square** is that which is both equilateral and right-angled; an **oblong** [rectangle] that which is right-angled but not equilateral; a **rhombus** that which is equilateral but not right-angled; and a **rhomboid** [parallelogram] that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. And let quadrilaterals other than these be called **trapezia**.
23. **Parallel** straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

C.2 Postulates

Let the following be postulated:

1. To draw a straight line from any point to any point.
2. To produce a finite straight line continuously in a straight line.
3. To describe a circle with any centre and distance.
4. That all right angles are equal to one another.
5. That, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.

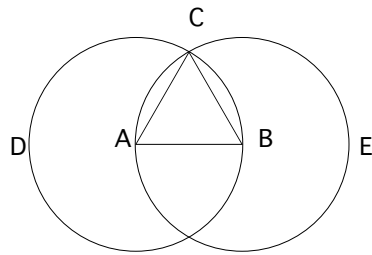
C.3 Common Notions

1. Things which are equal to the same thing are also equal to one another.
2. If equals be added to equals, the wholes are equal.
3. If equals be subtracted from equals, the remainders are equal.
4. Things which coincide with one another are equal to one another.
5. The whole is greater than the part.

C.4 Propositions

Proposition 1.

On a given finite straight line to construct an equilateral triangle.



Let AB be the given finite straight line.

Thus it is required to construct an equilateral triangle on the straight line AB.

With centre A and distance AB let the circle BCD be described; [Post. 3] again, with centre B and distance BA let the circle ACE be described; [Post. 3] and from the point C, in which the circles cut one another, to the points A, B let the straight lines CA, CB be joined. [Post. 1]

Now, since the point A is the centre of the circle CDB, AC is equal to AB. [Def. 15]

Again, since the point B is the centre of the circle CAE, BC is equal to BA. [Def. 15]

But CA was also proved equal to AB; therefore each of the straight lines CA, CB is equal to AB.

And things which are equal to the same thing are also equal to one another; [C.N. 1] therefore CA is also equal to CB.

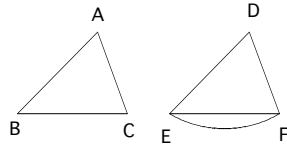
Therefore the three straight lines CA, AB, BC are equal to one another.

Therefore the triangle ABC is equilateral; and it has been constructed on the given finite straight line AB.

(Being) what it was required to do.

Proposition 4.

If two triangles have the two sides equal to two sides respectively, and have the angles contained by the equal straight lines equal, they will also have the base equal to the base, the triangle will be equal to the triangle, and the remaining angles will be equal to the remaining angles respectively, namely those which the equal sides subtend.



Let ABC , DEF be two triangles having the two sides AB , AC equal to the two sides DE , DF respectively, namely AB to DE and AC to DF , and the angle BAC equal to the angle EDF .

I say that the base BC is also equal to the base EF , the triangle ABC will be equal to the triangle DEF , and the remaining angles will be equal to the remaining angles respectively, namely those which the equal sides subtend, that is, the angle ABC to the angle DEF , and the angle ACB to the angle DFE .

For, if the triangle ABC be applied to the triangle DEF , and if the point A be placed on the point D and the straight line AB on DE , then the point B will also coincide with E , because AB is equal to DE .

Again, AB coinciding with DE , the straight line AC will also coincide with DF , because the angle BAC is equal to the angle EDF ; hence the point C will also coincide with the point F , because AC is again equal to DF .

But B also coincided with E ; hence the base BC will coincide with the base EF .

[For if, when B coincides with E and C with F , the base BC does not coincide with the base EF , two straight lines will enclose a space: which is impossible. Therefore the base BC will coincide with EF] and will be equal to it. [C.N. 4]

Thus the whole triangle ABC will coincide with the whole triangle DEF , and will be equal to it.

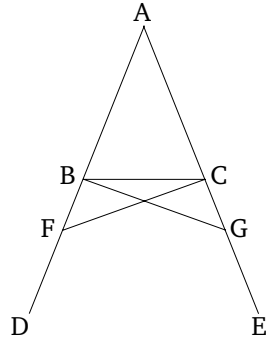
And the remaining angles will also coincide with the remaining angles and will be equal to them, the angle ABC to the angle DEF , and the angle ACB to the angle DFE .

Therefore etc.

(Being) what it was required to prove.

Proposition 5.

In isosceles triangles the angles at the base are equal to one another, and, if the equal straight lines be produced further, the angles under the base will be equal to one another.



Let ABC be an isosceles triangle having the side AB equal to the side AC; and let the straight lines BD, CE be produced further in a straight line with AB, AC. [Post. 2]

I say that the angle ABC is equal to the angle ACB, and the angle CBD to the angle BCE.

Let a point F be taken at random on BD; from AE the greater let AG be cut off equal to AF the less; [I. 3] and let the straight lines FC, GB be joined. [Post. 1]

Then, since AF is equal to AG and AB to AC, the two sides FA, AC are equal to the two sides GA, AB, respectively; and they contain a common angle, the angle FAG. Therefore the base FC is equal to the base GB, and the triangle AFC is equal to the triangle AGB, and the remaining angles will be equal to the remaining angles respectively, namely those which the equal sides subtend, that is, the angle ACF to the angle ABG, and the angle AFC to the angle AGB. [I. 4]

And, since the whole AF is equal to the whole AG, and in these AB is equal to AC, the remainder BF is equal to the remainder CG.

But FC was also proved equal to GB; therefore the two sides BF, FC are equal to the two sides CG, GB respectively; and the angle BFC is equal to the angle CGB, while the base BC is common to them; therefore the triangle BFC is also equal to the triangle CGB, and the remaining angles will be equal to the remaining angles respectively, namely those which the equal sides subtend; therefore the angle FBC is equal to the angle GCB, and the angle BCF to the angle CBG.

Accordingly, since the whole angle ABG was proved equal to the angle ACF, and in these the angle CBG is equal to the angle BCF, the remaining angle ABC is equal to the remaining angle ACB; and they are at the base of the triangle ABC. But the angle FBC was also proved equal to the angle GCB; and they are under the base.

Therefore etc.

Q. E. D.

Appendix D

Problems from *Experiencing Geometry*

These problems are taken from *Experiencing Geometry: Euclidean and non-Euclidean with History* [Henderson & Taimiņa, 2005].

Problem 1.1: When Do You Call a Line Straight?

- a. How can you check in a practical way if something is straight? How do you construct something straight—lay out fence posts in a straight line, or draw a straight line? Do this without assuming that you have a ruler, for then we will ask, “How can you check that the ruler is straight?”
- b. What symmetries does a straight line have?
- c. What is in common among the different notions of straightness? Can you write a definition of “straight line”?

Problem 2.1: What is Straight on a Sphere?

- a. Imagine yourself to be a bug crawling around on a sphere. (This bug can neither fly nor burrow into the sphere.) The bug’s universe is just the surface; it never leaves it. What is “straight” for this bug? What will the bug see or experience as straight? How can you convince yourself of this? Use the properties of straightness (such as symmetries) that you talked about in Problem 1.1.

Problem 3.1: What is an Angle?

Give some possible definitions of the term “angle.” Do all of these definitions apply to the plane as well as to spheres? What are the advantages and disadvantages of each?

For each definition, what does it mean for two angles to be congruent? How can we check?

Problem 3.2: Vertical Angle Theorem (VAT)

Prove: Opposite angles formed by two intersecting straight lines are congruent. (Note: [Such] angles ... are called **vertical angles**.) What properties of straight lines and/or the plane are you using in your proof? Does your proof also work on a sphere? Why? Which definitions from Problem 3.1 are you using in your proof?

Problem 7.1: The Area of a Triangle on a Sphere

- a. The two sides of each interior angle of a triangle Δ on a sphere determine two congruent lunes with lune angle the same as the interior angle. Show how the three pairs of lunes determined by the three interior angles α , β , γ , cover the sphere with some overlap. (What is the overlap?)
- b. Find a formula for the area of a lune with lune angle θ in terms of θ and the (surface) area of the sphere (of radius ρ), which you can call S_ρ . Use radian measure for angles.
- c. Find a formula for the area of a triangle on a sphere of radius ρ .

Teaching Notes: Several problems from *Experiencing Geometry: Euclidean and non-Euclidean with History* [Henderson & Taimiņa, 2005] discussed in this course are reproduced here. In the original book, each problem is given along with extensive commentary which students need to read in order to make sense of the problems. Students should therefore be asked to purchase this book or a course packet reproducing the relevant sections of the book. Most campus stores can arrange to obtain the rights to reproduce an excerpt of a book in a course packet.

Appendix E

Visualization on Cones and Pool Tables using Geometer's Sketchpad

In this section, the paper “Visualization on Cones and Pool Tables using Geometer’s Sketchpad” is reprinted from [Miller, 2006].

E.1 Introduction

In this paper, I will discuss two related activities that I have developed for use in a college geometry course directed at pre-service secondary and elementary school teachers. These activities ask students to develop geometry on pool tables and on cones. The two activities are closely related, and both ask students to relate computer models with physical models. The pool table activity was developed as a lead in for the cone project, which is by far the most successful project that I have used for getting pre-service teachers actively engaged with the full cycle of exploring a new space, making conjectures about it, and then coming up with explanations for and proofs of the conjectures.

The first activity is a computer lab using Geometer’s Sketchpad, in which students are asked to construct a model of a pool table, showing all of the paths that a cue ball can take before hitting another ball after bouncing off of a given number of rails. They do this by looking at all possible repeated reflections of the target ball over sides of the table. This leads naturally to a kind of covering space construction and a proof that there are $4n$ paths in which the ball hits exactly n rails. The other project comes later in the course, and asks students to write an open-ended paper in which they explore aspects of geometry on the cone on their own, in groups. One of the reasons that this assignment has been so fruitful is that there are many different directions in which students can take it, and several of these have related back to the pool table lab in ways that have sometimes surprised me. Students have used the ideas developed in the pool table lab to construct 2D covering space models of the cone in Sketchpad; printed and cut these out in order to make physical models of the cone; photocopied them onto transparencies to give a wonderful, physical representation of a covering space; and used their computer models to answer questions such as that of how many straight paths connect two points on a particular cone.

The main course in which I have used these activities has been a college geometry course at the University of Northern Colorado. Typically, half of the students in this class are math majors who intend to teach mathematics at the high school level, and half of the students are pre-service elementary school teachers who have chosen a concentration in mathematics. The course is taught in a modified Moore method, discovery-

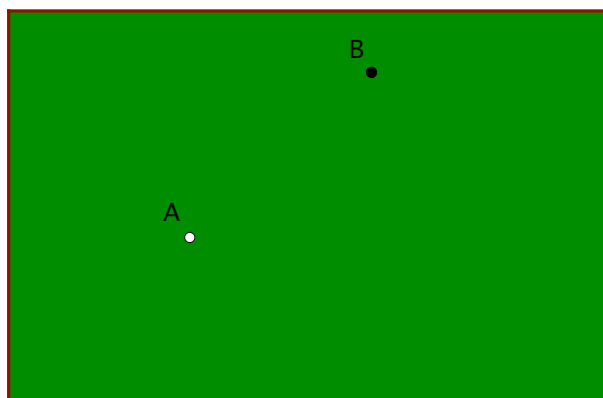


Figure E.1: A Sketchpad Pool table. How can we construct the path that the white cue ball A will take after being hit in the direction of point B?

learning centered style. It includes a great deal of writing, and is loosely based around the materials in chapters 1–4, 6, and 7 in the book *Experiencing Geometry: Euclidean and non-Euclidean with History* [2], although the course has no official textbook that the students must buy. Most of the materials used in the course are available on the web at <http://www.mathsci.unco.edu/course/math341/>.

The two activities described in this paper take up a significant portion of this course: the pool table activity described in section E.2 typically takes a week of class time to complete, and the cone project described in section E.3 typically takes three weeks of class time. I have found that the investment of time is well worth it. These activities are deeply connected to and give students significant experience with three of the four NCTM content standards for geometry, which state that students should learn to:

- analyze characteristics and properties of two- and three-dimensional geometric shapes and develop mathematical arguments about geometric relationships;
- apply transformations and use symmetry to analyze mathematical situations; and
- use visualization, spatial reasoning, and geometric modeling to solve problems.

Just as importantly, these activities give students an unparalleled opportunity to make mathematical discoveries themselves, to develop their own ideas, and to communicate these ideas to their peers. Thus, they also give students an extended experience with the NCTM problem solving, reasoning and proof, and communications process standards.

E.2 Geometry on Pool Tables

The first activity that I want to discuss asks students to look at geometry on a pool table. It asks them to find a way to predict the path a ball will take after bouncing off of a rail, using Geometer's Sketchpad; and then to use this to figure out what direction to aim a ball to hit another ball after one bounce off of a rail; after 2 bounces; and after 3 bounces. Finally, students are asked to determine a formula $f(n)$ for the number of different paths that a cue ball can take before hitting another ball after n bounces.

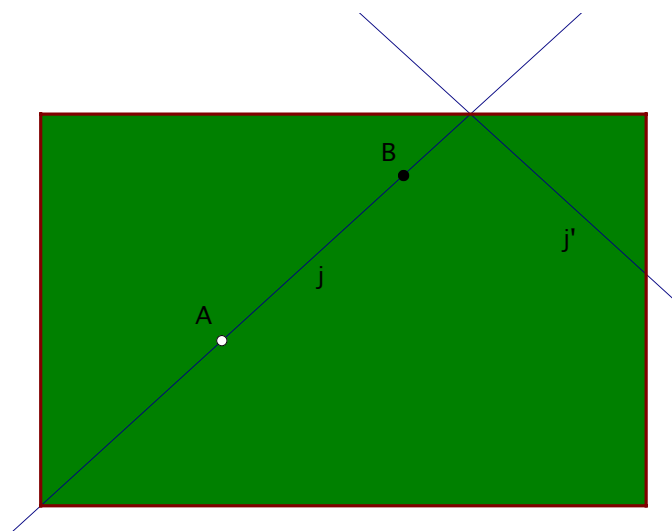


Figure E.2: j' is the reflection of j over the top rail of the table, and shows the path the ball takes after bouncing.

Figure E.1 shows a pool table constructed in Geometer's Sketchpad. The picture reproduced here is in black and white, but students seem to derive a surprising amount of pleasure from making their constructed tables green, so that they look more like real pool tables. The first problem that we want to consider is how to construct the path that the white cue ball A will take when hit towards another point B , and, in particular, the path it follows after it bounces against the top rail.

There are several different ways we could solve this problem, but the most useful way to do it is also one of the least obvious: draw the line connecting the two balls, and then reflect that line over the top rail, as shown in figure E.2, in which line j' is the reflection of line j over the top rail. Students can show that this makes the angle of incidence congruent to the angle of reflection; this follows from the vertical angle theorem and the fact that reflections preserve angle measure.

Now, how can we aim the cue ball so that it bounces off of the top rail *before* hitting point B , which we now think of as representing the 8-ball? The trick is to use what we just figured out: if we want the line showing the initial path of the cue ball to hit B after it is reflected over the rail, we should aim it for the reflection B' of B over this rail, as illustrated in figure E.3. This method of aiming for a reflected ball can be found in some books on playing pool. ([3], for example.)

We can extend this idea further, to paths that bounce more than once. For example, figure E.4 shows a path in which the cue ball bounces twice before hitting the eight ball, constructed by aiming for the point B'' that was created by reflecting the point B' over one of the other sides of the table. The initial path, when reflected over that side, becomes a path aiming for B' , which when reflected over the top rail becomes a path aiming for B . We can extend this idea to create paths that bounce any given number of times by reflecting the target ball that many times over the sides of the table.

Now that we know how to create paths that bounce any given number of times, we can ask the question: how many different paths are there that bounce n times. Many students initially guess that there should be $4 \cdot 3^{(n-1)}$ paths, because there are four choices for the first wall to bounce off of, and then three choices for each subsequent bounce, because the ball

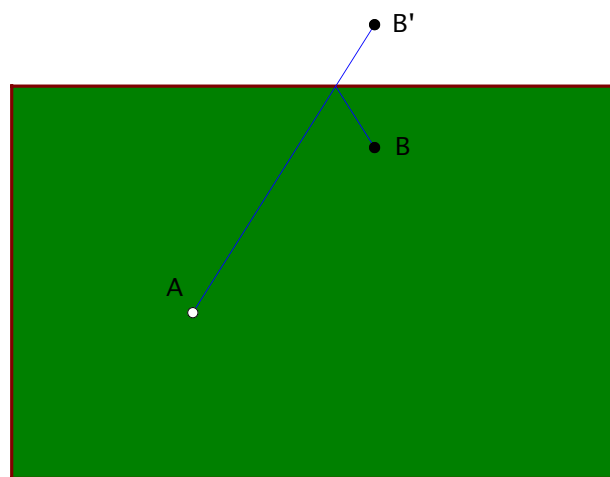


Figure E.3: To construct a path that hits B after one bounce, aim for B' .

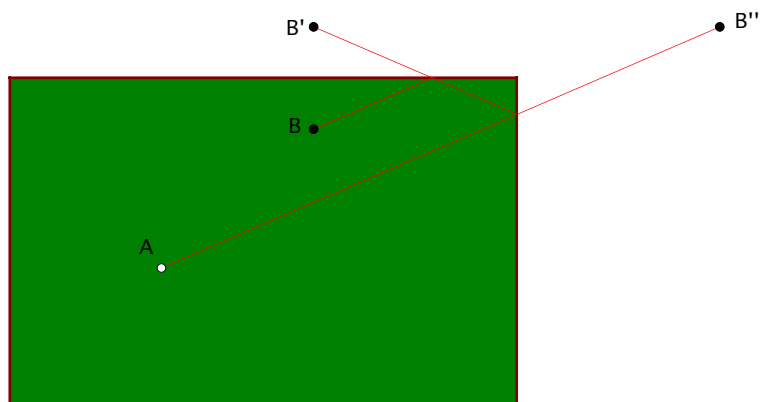
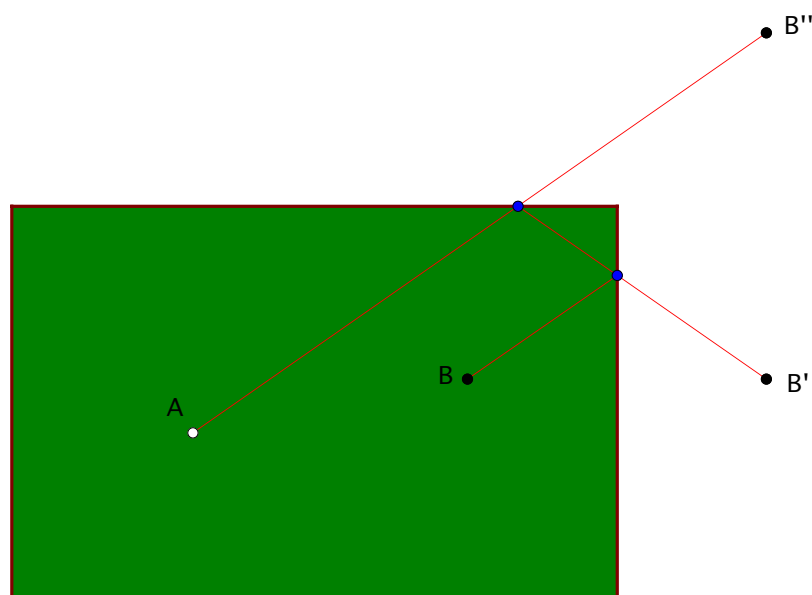


Figure E.4: To construct a path that hits B after two bounces, aim for B'' .

Figure E.5: Another path that hits B after two bounces.

cannot bounce against the same wall twice in a row. This does give an upper bound for the possible number of paths, but it is too big: not all possible sequences of sides correspond to actual paths, depending on the locations of the balls. For example, in figure E.4, in the path shown, the cue ball bounces off of the right rail and then the top rail; but there isn't any path that bounces off of the top rail and then the right rail. We can move the eight ball so that there will be a path that bounces off of the top rail and then the right rail, as shown in figure E.5, but then there won't be a path that bounces off of the right rail and then the top rail. Notice, though, that this path is still aiming for the point B'' obtained as before by two reflections of the point B . This same point can be obtained in two different ways, by reflecting over the top and right sides of the table in either possible order. Only one of these ways of getting B'' will correspond to an actual path, though. Which way it is will depend on where the balls are, and which wall the line connecting the cue ball to B'' crosses first.

This suggests a better way to count the number of possible paths with n bounces: count the image balls obtained by reflecting B n times over different walls. This idea gives rise to the grid of reflected pool tables shown in figure E.6. Our original table is in the center. There are four tables around the original table obtained by reflecting it over its four sides. This shows that there are exactly four different ways of aiming the cue ball to hit the eight ball with one bounce: we can aim for the reflection of the eight ball on any of these four tables. Likewise, there are eight tables that can be obtained from our original table by performing two reflections; they form a diamond shape around the previous four tables. These eight tables give us exactly eight different ways of hitting the eight ball after two bounces, again by aiming for the eight doubly reflected images of the eight ball. This pattern continues: there are twelve tables obtained by reflecting the original table three times, again arranged in a diamond formation around the previous eight, so there are twelve possible paths that bounce three times. In general, each diamond will contain four more tables than the previous one, so there must always be $4n$ paths that reach the eight ball after bouncing n times. (We are ignoring the possible problems of pockets, corners, and balls in the way.)

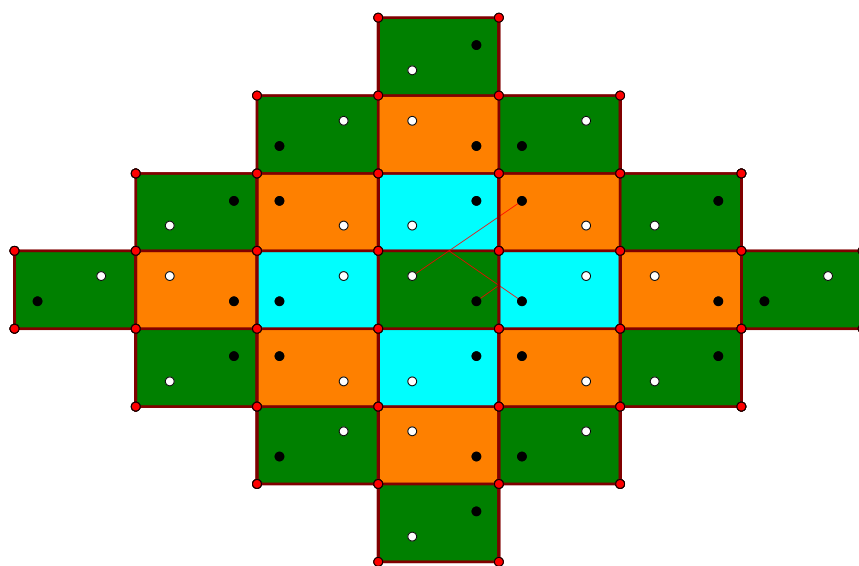


Figure E.6: The grid of reflected pool tables.

In my class, we generally spend between two and three class periods, each an hour and fifteen minutes long, on this problem. In that time, in my experience, most students (with varying degrees of help from me) can figure out how to construct paths with up to three bounces, and some but not all of them will have figured out how to count the number of possible paths with n bounces.

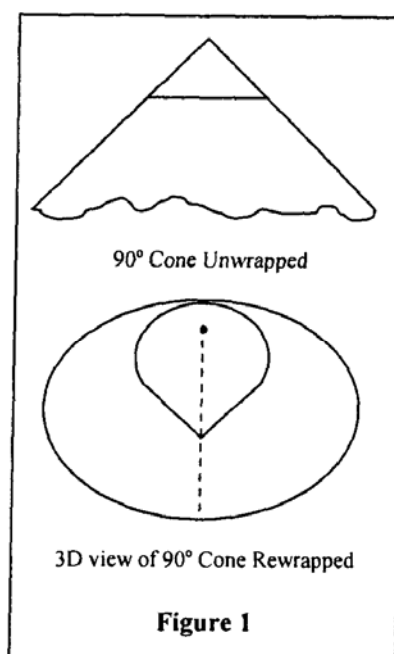
It should be noted that the pool table, with lines given by the paths of the balls, is not quite a manifold, because it doesn't look like the plane at the edges; but what we have given is essentially a covering space construction.

E.3 Cone Project

The preceding pool table project is fun, and the students enjoy it because it connects to a real world problem that they are interested in. However, it also serves as a perfect lead in to our final project.

This final project in the class asks students to explore geometry on the cone in groups of three. The project is very open-ended: students explore what interests them, write a report, and present their findings to their classmates. We spend a lot of time in class on this project—generally, three to four weeks. I have found that it has always been worth the investment in time; it has been the best project I have used for getting students at this level really doing mathematics themselves. It is based on the problems in chapter four of [2], but its approach is much more open ended, so that students feel like they are picking their own problems to explore in a subject they haven't been told anything about in advance—*i.e.*, like they are mathematicians themselves.

In order for this assignment to make sense, I need to explain a little bit about geometry on the cone. The cones that we want to consider are infinite cones, that extend infinitely without a base. These cones have one special point, the tip, called the *cone point*, where the surface is not smooth. If we cut the cone open along a straight ray coming out of the

Figure E.7: Illustration of a cone with a 90° cone angle.

cone point, we get the shape of an angle that we can lay flat; and we can measure the size of the cone by the size of this angle, called the *cone angle*. See figure E.7 for an illustration of this idea, taken from a student paper [1]. Notice that a straight line on the flattened cone becomes a straight line on unflattened cone when the sides of the angle are glued back together. Lines on the cone that do not pass through the cone point are straight if they are straight when the cone is flattened; and we will consider lines that pass through the cone point to be straight if they have half-turn symmetry—that is, if the two rays coming out of the cone point make an angle that is exactly half of the cone angle. Lines like these can be thought of as being the “straightest possible” lines through the cone point.

Figure E.8 shows the text of the cone assignment as given to the students. It consists mainly of a list of questions that the students can choose to investigate. Some students have written successful papers that mainly investigated a single question, while others have looked at several or even most of these questions. Some of the questions refer to the idea of holonomy, which may be an unfamiliar concept for many readers; holonomy won’t be discussed further in this paper, but the interested reader can find its definition in chapter 7 of [2].

Consider two points A and B on a cone with a 90° cone angle—that is, on a cone that can be sliced open and laid flat to form a 90° angle, as shown in figure E.9. We’d like to understand how to find the line that connects them on this cone. We can easily draw a line k between the two points, as shown in figure E.10, but we need to figure out what happens to the line after it hits one of the sides of our flattened cone. Notice how similar this is to the previous problem of predicting the path that a ball would take on a pool table after bouncing against a wall. In fact, we can imagine that our points on the cone are two dimensional balls in a two dimensional flatlander’s game of pool on a cone—but without any walls to bounce off of. If we hit ball A towards ball B , ball B will continue along the

Geometry on the Cone

See how much geometry you can develop on cones with cone angle < 360 degrees (e.g., the 90 degree cone). You may also be interested in comparing these with cones with cone angle > 360 degrees (e.g., the 450 degree cone) and/or cylinders.

For any of these surfaces you can ask any of the following questions:

- What do the intrinsically straight lines look like? What happens to lines that run into the cone point?
- What do circles look like?
- Which of Euclid's Postulates are true?
- What is the sum of the interior angles of a triangle?
- What is the holonomy of a triangle?
- Which triangle congruence theorems hold (or don't hold)?
- Is there a unique straight line joining any two points? If not, for which points is there a unique straight line joining them?
- Do any straight lines intersect themselves? If so, how many times?

Explore any of these questions, or others, that interest you, and write a paper explaining what you find.

Figure E.8: Text of the cone project assignment given to students.

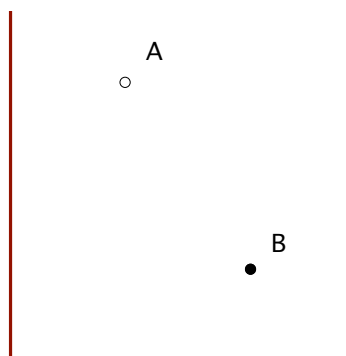


Figure E.9: Two points on a flattened 90° cone.

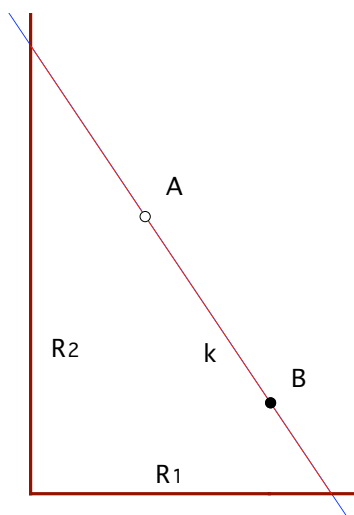


Figure E.10: What happens if we extend line k on the cone?

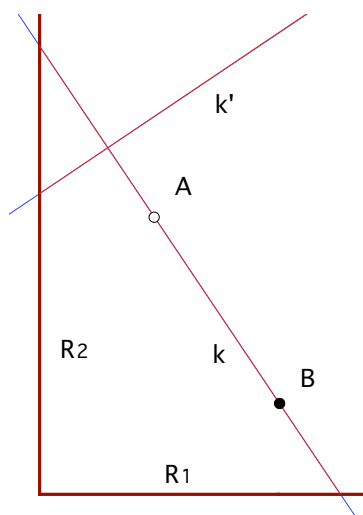
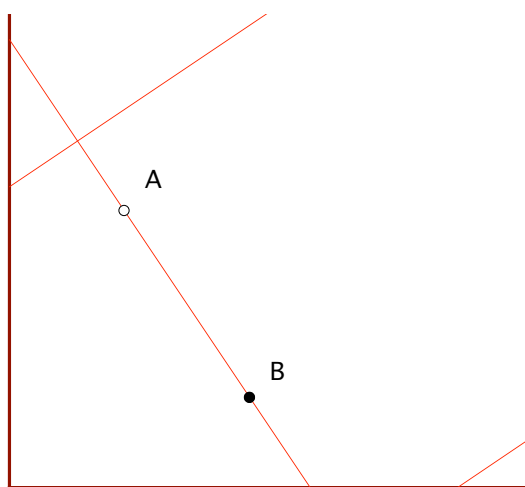


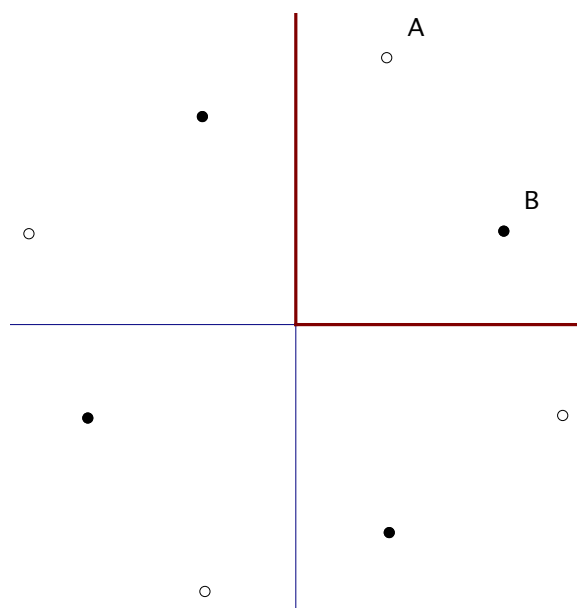
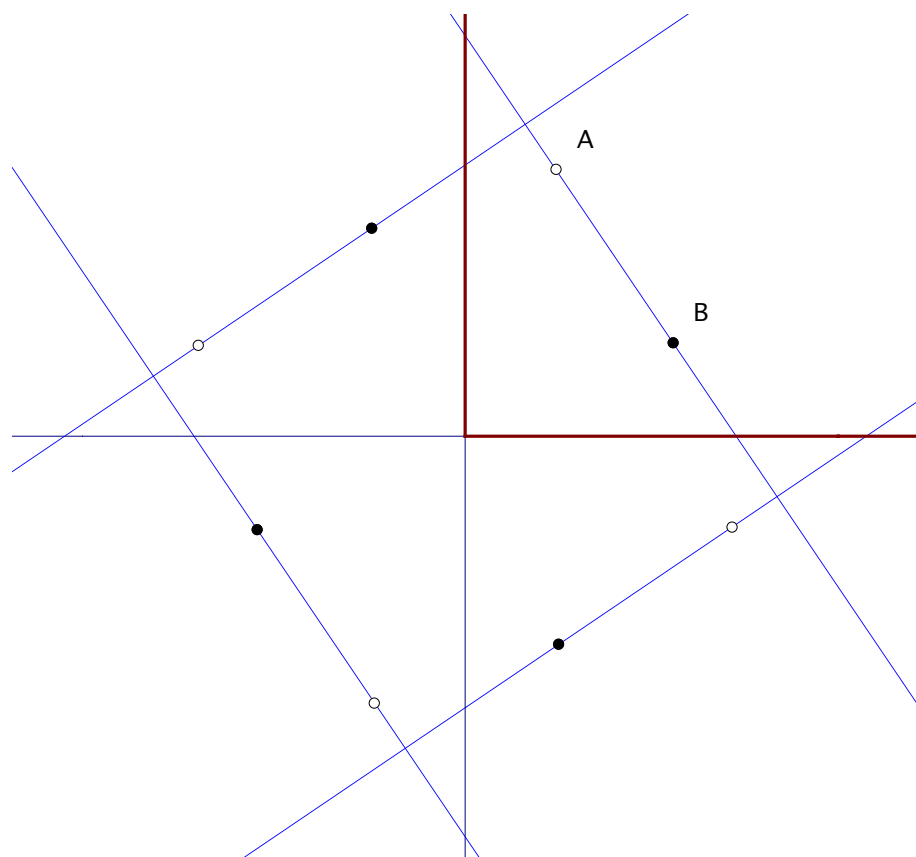
Figure E.11: The line k' is formed by rotating k by 90° about the cone point.

Figure E.12: A complete line on the 90° cone.

same line until it reaches the bottom-right side of our flattened cone, labeled R_1 . Since the cone is made by gluing R_1 to R_2 , the path must continue from a point on R_2 that is the same distance from the cone point as the point where it hit R_1 —we'll call the continuation of the path k' . Furthermore, the path must make the same angle with R_2 that it made with R_1 . Since R_1 and R_2 form a 90° angle, this means that k' must be rotated 90° from k . So the easiest way of finding the continuation of the path k is by rotating it by 90° about the cone point, as shown in figure E.11. If we use this method of rotation to extend the line in both directions, and then erase the parts of the lines that don't lie within our flattened cone, we get a picture of a line on the cone like the one shown in figure E.12. Students can easily figure out how to make pictures like this using Geometer's Sketchpad, and can then cut them out and tape them together to make really nice three dimensional cones. My experience is that students in my class always start out by making paper models and drawing figures on them by hand, but once one group discovers that they can make more accurate models using Sketchpad, then several other groups will notice and also start using Sketchpad.

Once we know how to draw lines on cones, we can ask the question of how many different lines can be drawn between two given points. This is analogous to the question on the pool table of how many different ways the cue ball can be aimed to hit the eight ball after bouncing a given number of times. In that case, we aimed for reflected copies of our original balls, because the continuation of a line was produced by reflecting it. In the present case, the continuation of a line is produced by rotating it about the cone point. This suggests that we rotate our original points about the cone point and aim for the rotated points. This leads us to the grid of points shown in figure E.13. Notice that we have again formed a covering space. Because there are four copies of the black point B in our grid, there are four different ways to connect the white point to the black point on our 90° cone: the direct way that we already found, shown in figure E.14; one way directly across the back of the cone, shown in figure E.15; and two paths that wrap around the cone and cross themselves before hitting the other point, shown in figures E.16 and E.17, one in each direction.

This should give the flavor of how this problem is related to the previous one, and how students can use these kinds of ideas to start exploring the properties of lines on the cone. Students can (and will, given the chance) explore many related questions using these kinds

Figure E.13: A grid of 90° cones.Figure E.14: How to connect two points on the 90° cone: Method 1.

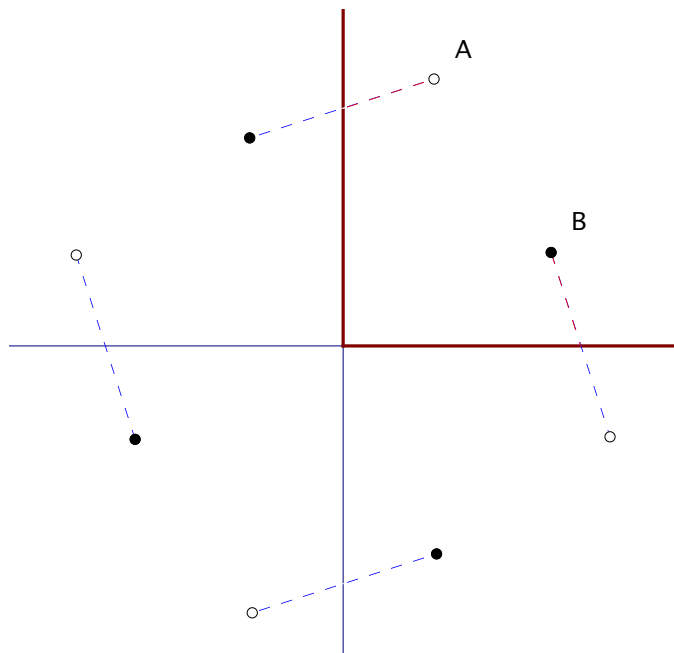


Figure E.15: How to connect two points on the 90° cone: Method 2.

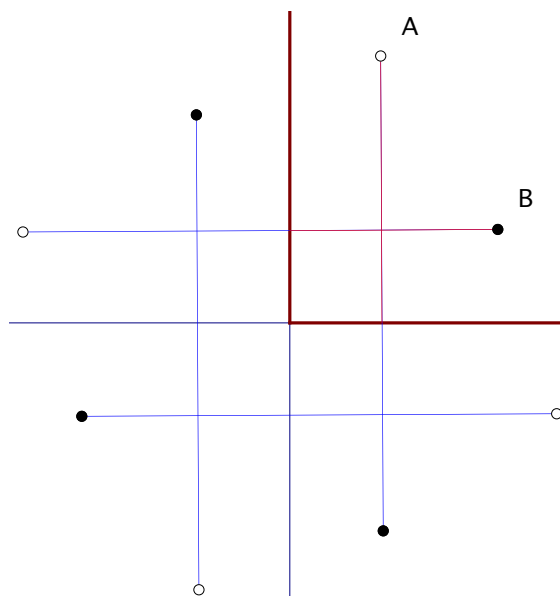


Figure E.16: How to connect two points on the 90° cone: Method 3.

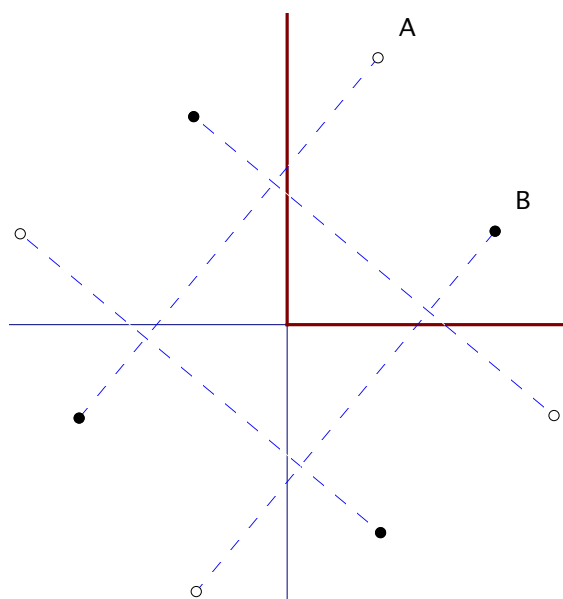


Figure E.17: How to connect two points on the 90° cone: Method 4.

of methods: How many ways are there of connecting two points on a cone? How many times will a line cross itself? At what angles? Does this depend on where the points or lines are, or what the cone angle is? How?

These questions can all be explored either by hand on paper models, or using Sketchpad; what's more, once correct conjectures have been made, they almost all have accessible proofs that some students will find.

One question that is not addressed in [2] and that has turned out to be much more interesting than I expected it to be when I first asked it is: What do circles look like on cones? This is a question that is much easier to explore using Sketchpad than it is to explore by hand. It also turns out to be a question that students really like.

Most students classify circles on the cone into 3 sets:

1. Those whose center is at the cone point. These are the circles formed by the intersection of the cone with a horizontal plane; they are circles both extrinsically and intrinsically, although intrinsically, they contain fewer than 360 degrees.
2. Those whose center is far away from the cone point, so that the cone point lies outside the circle. These look exactly like planar circles, for the most part, so they aren't that interesting, although occasionally students will notice that it is possible for them to intersect themselves, and all of the intersections lie along a single line through the cone point.
3. Those whose center is close to, but not on, the cone point. These are the most interesting ones. If we start with a circle like the one shown in figure E.18, we can use the covering space idea to draw rotated copies of the circle, as shown in figure E.19. If we cut out one quadrant of this figure and tape the sides together, we will get a perfect model of a circle on a cone. I have had students that have drawn circles like these by hand, using string or a compass, but they're much easier to produce using Sketchpad.

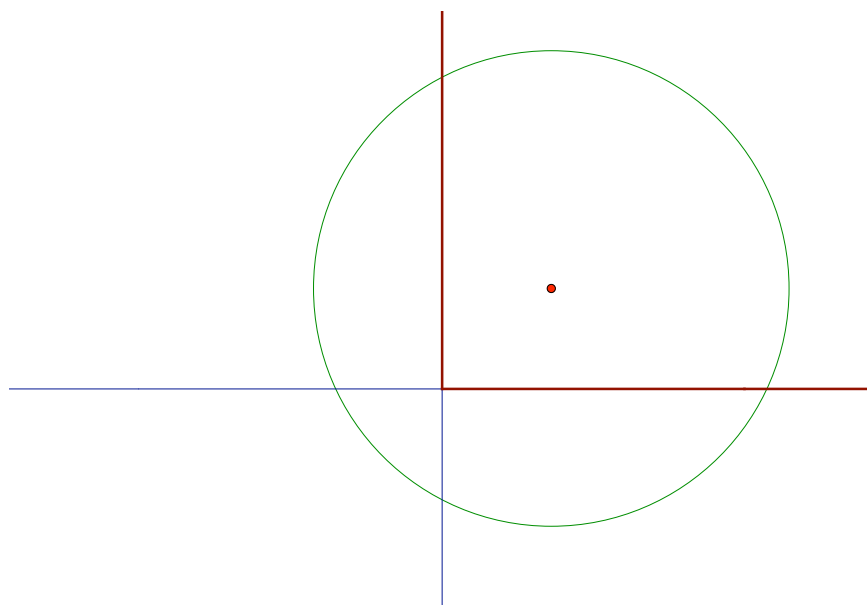


Figure E.18: What happens to this circle on the 90° cone?

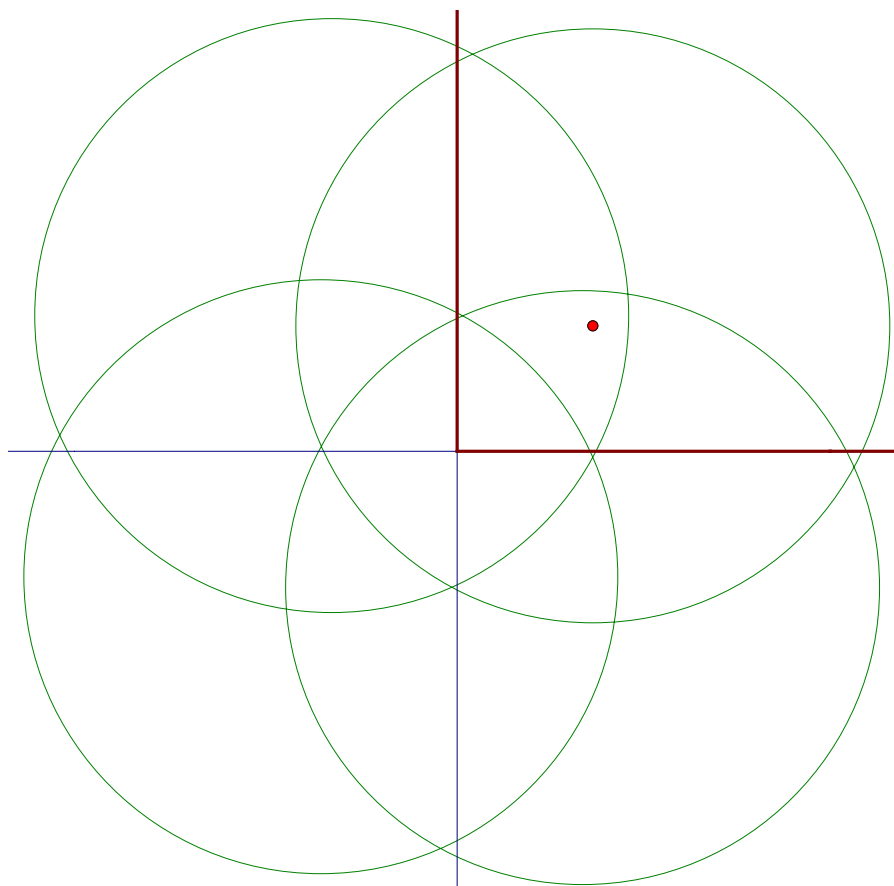


Figure E.19: A circle on the 90° cone.

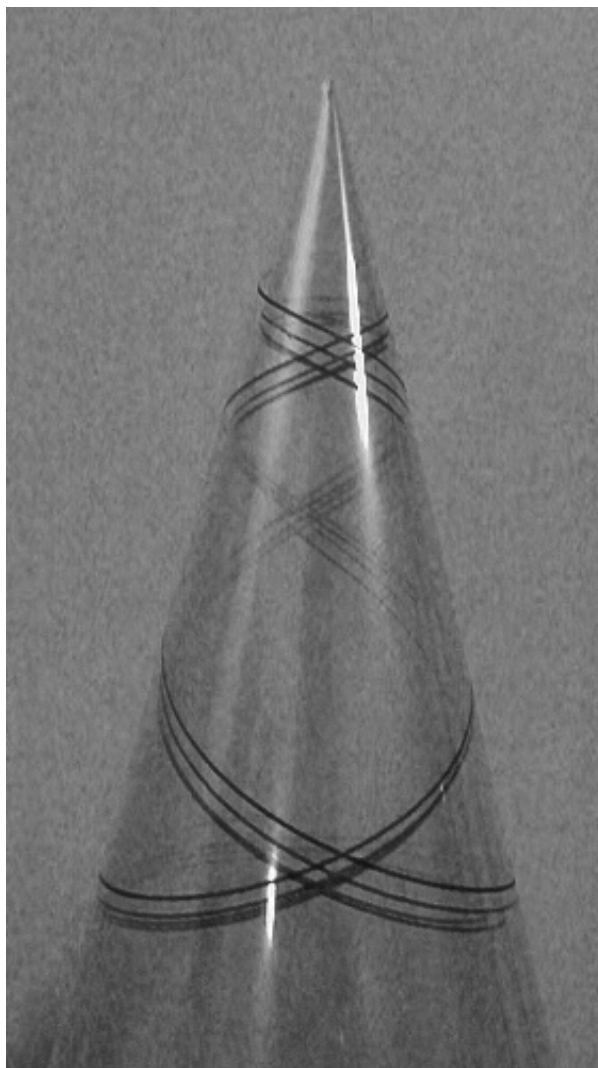


Figure E.20: A student-produced covering space model of a circle on a 90° cone, made by photocopying a picture like that shown in figure E.19 onto a transparency.

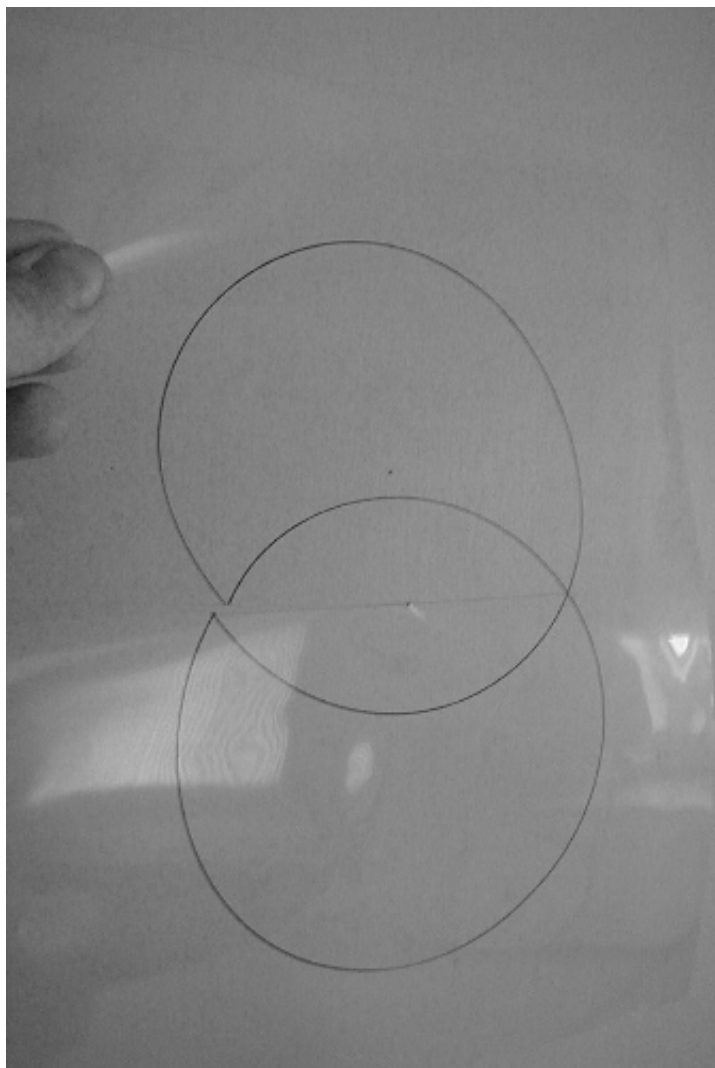


Figure E.21: A student-produced covering space model of a circle on a 180° cone, unwrapped.

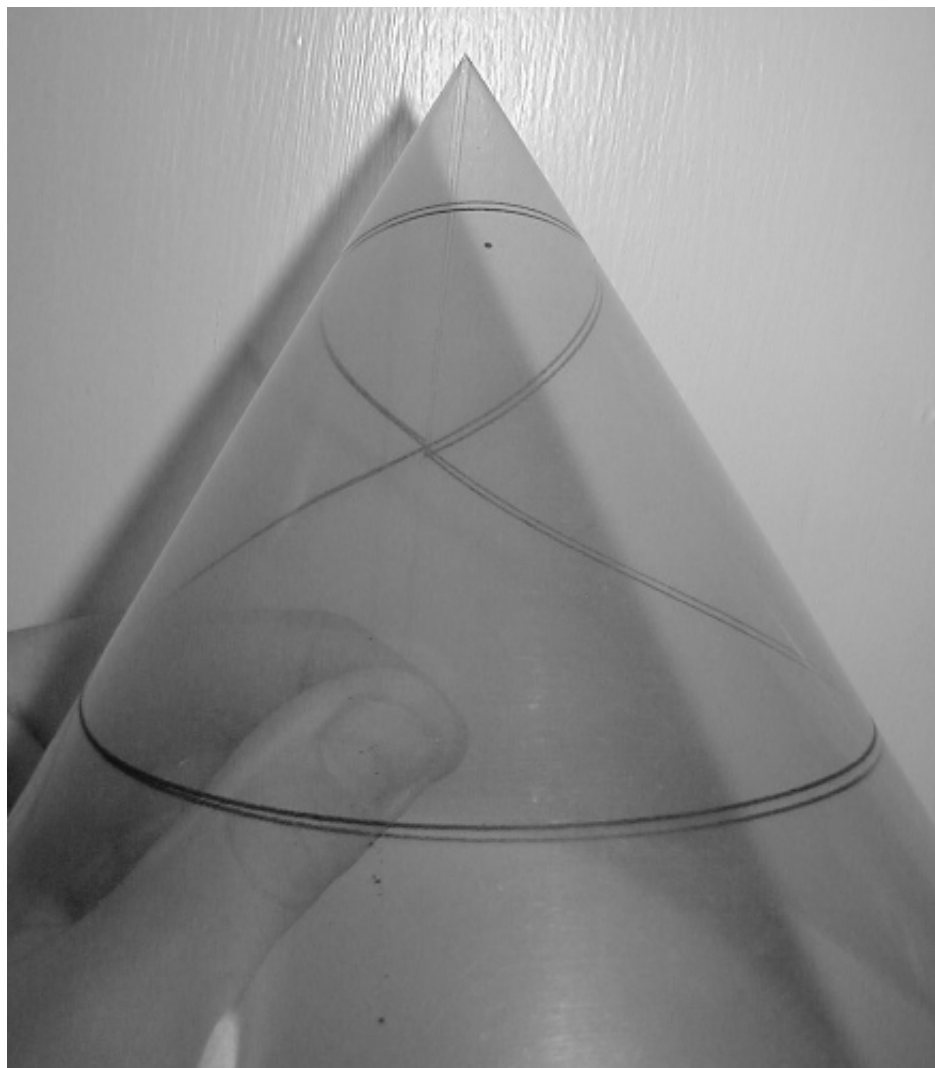


Figure E.22: A student-produced covering space model of a circle on a cone, made by wrapping up the transparency in figure E.21.

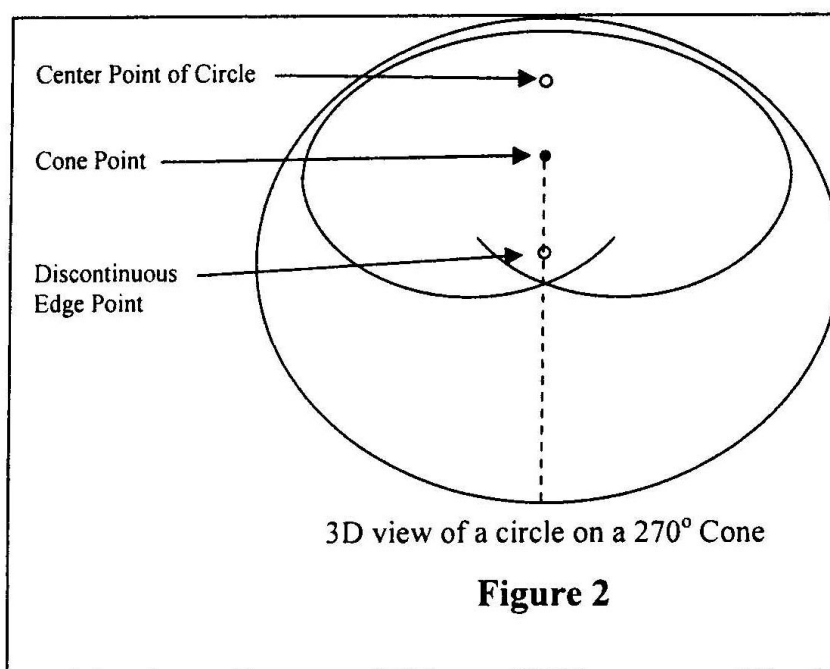


Figure E.23: A circle on a cone whose cone angle doesn't divide evenly into 360° : Version 1.

One particularly nice illustration of the idea of a covering space and how to use it to draw circles was invented by a group of pre-service elementary school teachers in my class last semester [5]. They came up with idea of photocopying Sketchpad sketches like the one in figure E.19 onto overhead transparencies, for use in their in-class presentation. One of their models of a circle on a 90° cone is shown in figure E.20. As another example, figure E.21 shows one of the other transparencies that they made to show a circle on the 180° cone, and figure E.22 shows the same transparency rolled up into a cone. I think that models like this would be really useful to show in any class discussing covering spaces for the first time.

A further question is: what do circles look like on cones whose cone angle does not divide evenly into 360° ? In this case, the plane isn't a covering space. Most students that have looked at this question have come up with an answer like that shown in figure E.23, which is again taken from a student paper [1]. We can consider a circle to be the set of all points that can be connected to a given center by a straight line segment of a given length. A radius of the circle can therefore be drawn in any direction from the center point; because there are 360 degrees around this point, there must be 360° worth of circle. In this particular case, we have a 270° cone, so we have 270° worth of circle before the circle intersects itself on the side of the cone directly opposite the center of the circle; since there are 90° of circle left over, we have 45° more on each side. There is one isolated point of the circle that isn't connected to the others, which is created by the radius that goes directly through the cone point.

Last semester, however, one group looking at this question came and showed me the picture in figure E.24, which they said was a picture of a circle on a flattened 260° cone, and told me that they had conjectured that the number of arcs in a circle like this was the

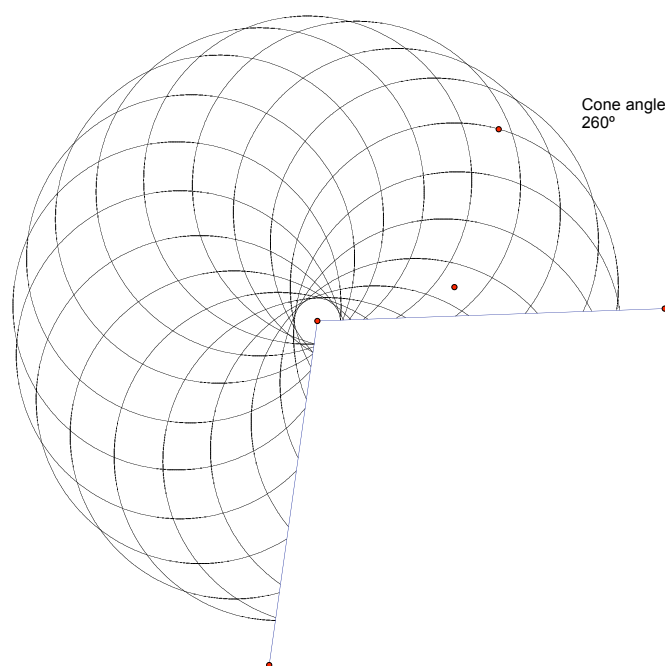


Figure E.24: A circle on a cone whose cone angle doesn't divide evenly into 360° : Version 2.

least common multiple of the cone angle and 360° , divided by the cone angle [4]. I told them that I didn't think this was right, having only seen examples like figure E.23 before. They persisted in saying that their example should be a circle, however, and we came to the conclusion that this could be a circle if you considered a circle to be what the students called "a continuous arc." In standard terminology, we would call this a curve of constant curvature—and, sure enough, their example is a curve of constant curvature—just like a circle on the plane. So which shape we consider to be a circle on the cone depends on which of the possible planar definitions of a circle we choose to generalize. These constant curvature circles are especially fun to look at on physical models of the cone; the way they match up feels surprising and cool.

There are a lot of reasons why I love doing this project with students: it builds on everything we have done all semester, and connects directly back to other assignments like the pool table assignment. It far more open ended than most undergraduate projects, and students can find fruitful things to explore in many different directions (only some of which have been discussed here!) They get to make discoveries and conjectures on their own in a new area, which is an experience I think we need to give our students as much as possible; and then they can often prove their own conjectures. But one aspect that I particularly enjoy is that every semester, students show me new ideas that I have never seen before, like the transparency cone models and the constant-curvature cone circles that students showed me last semester.

The assignment is well summed up by the conclusion of one group's paper from last spring: "We had a great time with this project because of how much constructive learning took place. We were surprised at how far we were able to develop our thoughts and build off each others ideas." [6] I'm continually surprised by this myself, and I think that more instructors should try giving this kind of assignment—they, too, might be surprised at how

much real mathematics their students are capable of!

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Appendix F

Midterm

In completing this exam, you may consult your notes and handouts from this class, but you may not consult any other sources or discuss the problems with anyone other than me.

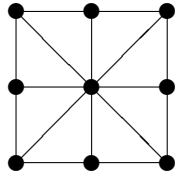
Your final copy will not be accepted if it does not include the following signed honor pledge:

I pledge on my honor that this examination represents my own work in accordance with University and class rules; I have not consulted any outside sources, nor have I given or received help in completing this exam.

1. A recent article in the New York Times, “Universe as Doughnut: New Data, New Debate” [Overbye, 2003] recounts new evidence that the physical universe may be a shape called a flat 3-torus. The 2-dimensional analogue of the shape is called a flat 2-torus, or just a flat torus. The flat torus is familiar from many old arcade games like asteroids, pac-man, and Spacewar. It is the shape formed from a rectangular piece of a plane by gluing the top of the rectangle to the bottom and the right side of the rectangle to the left side.

Explain which of Euclid’s postulates and of axioms 1–4’ from the worksheets are true on the flat torus.

2. Write a paragraph on each of the following:
 - (a) Here are some things that I think are going well in the course so far—let’s keep doing them:
 - (b) Here are some suggestions for the rest of the semester and future semesters:
 - (c) Here are some personal reflections on my work in this class so far:
3. Let NG be the axiom system described in the worksheets, consisting of axioms 1-4’.
 - (a) Which of the following are models of NG? For each potential model, explain why it is or is not a model of NG.
 - i. The northern hemisphere of a sphere, including the equator, in which the lines are the pieces of great circles that lie in that hemisphere.
 - ii. The southern hemisphere of a sphere, not including the equator, in which the lines are the pieces of great circles that lie in that hemisphere.
 - iii. The world drawn below, which contains nine points and eight lines.



- (b) Which of the following statements are theorems of NG? Prove each theorem, and for each non-theorem find a counterexample model of NG in which it is false.
- There exist four non-collinear points.
 - If A is a point, then there exist two other different points B and C such that A , B , and C are collinear.
 - If A is a point, then there exist two other different points B and C such that A , B , and C are non-collinear.
- (c) Let NG' be the axiom system in which axiom 1 has been replaced by axiom 1', as follows: Is the following statement a theorem of NG' ? If it is, prove it; if it isn't, then give a counterexample showing that it isn't.
- If A is a point, then there are at least three different lines through A .

Teaching Notes:

This is one of many different versions of the take home midterm I give. I generally give students a week to work on it on their own. They turn it in and I give it a numerical grade. After this, we talk about it in class, and then I give them the opportunity to revise it to get up to half of the points they lost back. This is probably the hardest assignment of the semester. The article referred to in Problem 1 can currently be downloaded for free from the New York Times website. (Search for the title of the article, and make sure that you are searching the whole New York Times archive, not just the last year.) I usually photocopy the article and hand it out along with the exam if I ask this question.

Other possible surfaces for Problem 1 are finite cylinders (with or without a top and bottom), infinite cylinders, and cubes.

Appendix G

The Story of What We Did This Semester

Our central question this semester has been: *How can we decide what is true in geometry?*

Because geometry is the study of “Earth Measurement,” we started by trying to measure how much of a wall could be seen through a viewing tube at a given distance. We started by gathering data about this situation, which led us to some conjectures about a possible pattern. We realized, however, that in order to determine if this pattern *always* held, we would need to give a convincing argument about *why* it always held. This led us to the idea of a *proof*: a convincing argument that shows that something always holds.

We next started examining geometry on the sphere. This also fit with the idea of “Earth Measurement,” since the Earth is indeed a sphere. It also provided us with a good venue for exploring how to decide what was true, since very few people in the class had ever studied spherical geometry or thought about this before.

We realized that in order to do any geometry on the sphere (or any new surface), we first have to decide what lines we are going to use as our straight lines, since most of the shapes we study in geometry (such as triangles, squares, or parallel lines) are built out of straight lines. This was uncomfortable for many people at first, since they were not used to creating definitions. Having read *Flatland*, we decided that the straight lines would be those that a Flatlander would experience as being intrinsically straight. We tried to imagine what a Flatlander’s experience would be like on the sphere, and tried to use this to write a general definition of what it means to be straight on any surface. In writing this definition, we tried to use properties that were objectively checkable by an outside evaluator. We tried to write definitions that were complete enough to exclude everything we didn’t consider to be straight, but that didn’t have some many pieces that they were too hard to check.

An important tool we used in writing these definitions was the notion of a *symmetry*, a rigid motion of a figure that brings it back on top of its starting position. Straight lines have many intrinsic symmetries, and many people used these in writing their definitions. Later in the course, we used Geometer’s Sketchpad, a dynamic geometry program, to explore the symmetries of triangles and quadrilaterals and to construct the path of a ball on a billiards table using symmetries.

Once we had settled on our definitions, we tried to decide what facts were true. We did this on the sphere, as well as other surfaces like the plane and the cone. In each case, we found that we were free to adopt any definitions we liked, but that this would then determine what facts were true.

In studying planar geometry, we turned to Euclid’s *Elements*, a work that dates back to 300 B.C. In the *Elements*, Euclid sets down a collection of definitions and assumptions, called *Postulates* and *Common Notions*, and then tries to carefully prove facts about planar

geometry from them. We looked at Euclid's Postulates on the sphere, and decided which of them were true there. We found, surprisingly, that all five of Euclid's Postulates were true on the sphere. We decided that this meant that if Euclid really only used those Postulates as his assumptions in his proofs, then all of his proofs should be equally valid on the sphere. However, we found that this was not the case; for example, we found that Euclid's Proposition Four about Side-Angle-Side triangle congruence is not true on the sphere. We concluded that some additional assumption(s) must have snuck into Euclid's proof. In this particular case, we were able to identify the additional assumption as Postulate 1': Two points determine a unique line. We were also able to find ways to restate Proposition four so that it was true on the sphere, and to modify Euclid's proof in order to prove our restated version without relying on Postulate 1'.

In order to better understand proofs in geometry, we considered a very simple axiom system for a part of geometry known as Neutral Geometry. This allowed us to consider the relationship between axioms, proofs, and models. We found that with more axioms, we could prove more theorems, but they would apply to fewer models. This is because a statement is a theorem if it is true in every model of the axioms. As we add axioms, we eliminate the possible worlds in which the new axioms are not true.

Throughout the semester, we studied a large variety of different spaces: the plane, the sphere, cones, billiard tables, and other spaces. In each, we went through a similar process in trying to understand its properties. First, we would try to get a sense of the new space and try to understand what geometric shapes looked like in the space. Next, we would try to generate examples and come up with possible patterns, formulas, and conjectures about statements that were true or not true in the space. Finally, we would try to come up with convincing arguments that our conjectures were always true, and tried to write them up in ways that would convince our peers that they were absolutely correct. In the end, this process is at the heart of what being a mathematician is all about.

Teaching Notes:

This is a summary of the whole class that I sometimes give out at our last meeting to try to tie everything together.

Appendix H

Sample Class Schedule

The following is a sample schedule for a typical fifteen week semester, with classes meeting twice a week for 75 minutes each. This schedule leaves 7 full class periods for working on the cone projects; if some of the earlier projects take longer than their scheduled amount of time, this extra time can be absorbed by spending less time on the cone projects.

#	In- class activities	Assignments due
1	Class introductions; discussion of the philosophy of this class; begin viewing tubes assignment	
2	Continue group work on viewing tubes; begin straightness on the sphere	Email introduction to instructor; Reading from <i>Flatland</i>
3	Students present viewing tubes work; continue with straightness on the sphere	I1: Viewing Tubes; Reading: Problems 1.1 and 2.1 from <i>Experiencing Geometry</i>
4	Straightness on the Sphere	
5	Finish Straightness on Sphere: discuss reading from Euclid; do Sketchpad Tutorial	Reading: Euclid: Definitions, Postulates, Common Notions, and Proposition 1
6	Computer Lab I	F1: Straightness on the Sphere first draft
7	Computer Lab I	
8	Euclid's Postulates on the Sphere	
9	Finish Postulates on Sphere; Discuss Proposition 4	Reading: Euclid: Proposition 4; I2: Computer Lab 1
10	Discuss Proposition 5; Start NG worksheets	Reading: Euclid: Proposition 5; F1: Straightness on the Sphere final draft
11	NG worksheets	I3: Euclid's Postulates on the Sphere
12	NG worksheets	
13	NG worksheets	
14	Side Angle Side on the Sphere	

15	Finish Side Angle Side on the Sphere; start Euclid's Postulates on the 450° cone	I4: Worksheets
16	Euclid's Postulates on the 450° cone	
17	<i>Experiencing Geometry</i> Problem 7.1	F2: Side-Angle-Side on the Sphere first draft
18	Finish Problem 7.1; start Computer Lab II	I5: Euclid's Postulates on the 450° cone
19	Computer Lab II	I6: Henderson 7.1
20	Midterm given out; finish Computer Lab II	
21	Discuss holonomy; start final cone projects	
22	Cone projects; introduce NCTM standards assignment	F4: Midterm Due
23	Cone projects	F2: Side-Angle-Side on the Sphere final draft
24	Cone projects	I7: Computer Lab 2
25	Cone projects	
26	Discuss NCTM Standards; use any time left over for cone projects	I8: NCTM Standards
27	Cone projects	
28	Cone projects	F5: Cone Project Drafts Due
29	Watch shape of space videos; work on shape of space assignment	
30	Work on shape of space assignment and cone projects	
31	Final class meeting during finals week; students present final cone projects	Final cone projects; I9: Shape of Space; and completed portfolios

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